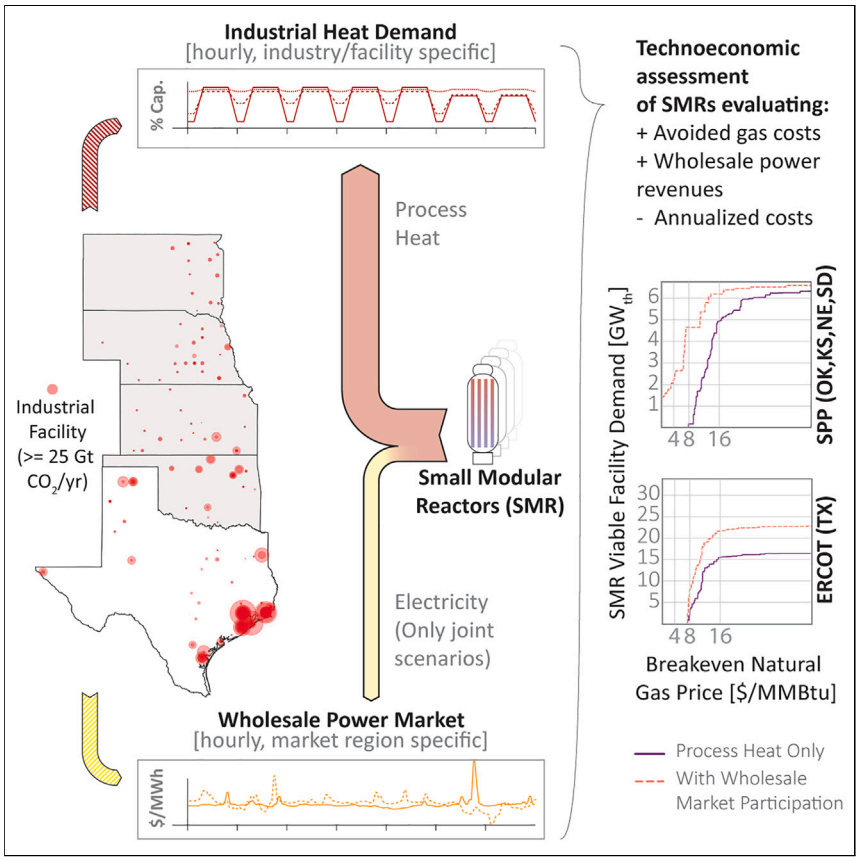
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Technoeconomic analysis of small modular reactors decarbonizing industrial process heat



We analyze the technological and economic viability of decarbonizing industry with nuclear small modular reactors. By examining five small modular reactor designs at 357 potential industrial facility processes across five US states, we find small modular reactors can economically displace natural gas use at 113 facility processes. Deploying reactors at a subset of these facilities could enable the scale up of this emerging low-carbon technology while contributing to industrial decarbonization.

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Highlights

Small modular nuclear reactors (SMRs) could help decarbonize high heat industry

At 2021 US natural gas prices, SMRs are not viable for process heat only

Providing process heat and wholesale electricity significantly improves SMR viability

Industrial process scale and heat quality requirements dictate economic viability

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Technoeconomic analysis of small modular reactors decarbonizing industrial process heat

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SUMMARY

Providing high-temperature industrial heat is a key decarbonization challenge. Nuclear small modular reactors (SMRs) could meet this decarbonization challenge, but no technoeconomic analyses of their potential exist. We quantify the technoeconomic potential of five SMR types for displacing natural gas providing process heat for 357 industrial facility processes across five US states. At 2021 US gas prices, we find no SMRs are economically viable when providing process heat alone. 28 SMR modules (5.6 GW_{th}) and 256 modules (28.6 GW_{th}) are economically viable at avoided gas costs of $8 and $16/MMBtu, respectively. By also participating in wholesale power markets, we find up to 113 industrial facility processes (6% of industrial demand) could be economically served by 33.9 GW_{th} of SMRs, avoiding 4 million tons of CO_2 emissions annually at 2021 US gas prices. Our analysis identifies key criteria that drive SMR economic viability, including thermal demand quantity and quality matching.

INTRODUCTION

Aggressively mitigating climate change requires reducing carbon dioxide (CO_2) emissions from all sectors, including a 65%–90% decrease in the industrial sector by 2050 to avoid warming greater than 1.5°C.[1] In the United States as of 2019, industry emitted 1,522 million metric tons of CO_2-equiv per year, or 23% of total national emissions.[2] In the same year, fossil fuel combustion accounted for 77% of US industrial thermal energy requirements[3] with roughly half of emissions coming from direct combustion of fossil fuels for process heat[2] in industrial processes ranging from less than 150°C to those above 1,000°C.[4] Low-carbon provision of high temperature process heat (>300°C) forms a core challenge to industrial decarbonization because there is currently no clear path for readily deployable and inexpensive alternatives to fossil fuel combustion.[5,6] Here, we focus on hard-to-decarbonize heat processes with temperatures ranging from 300°C to 850°C, which are common in a variety of industries, such as chemical and plastics manufacturing, food processing, and petroleum refining. The processes associated with this temperature range account for 40% of emissions from reporting facilities.[7] These processes may be decarbonized with several developing technologies, including electrification, green or blue hydrogen, ammonia, geothermal, concentrating solar power, and advanced small modular nuclear reactors.[4,5,8–11] Each of these technologies has research and implementation barriers, such as limited temperature ranges.[4,12] But only advanced small modular nuclear reactors are not limited by spatially variable resource availability and can be broadly deployed at existing industrial facilities.[4,13]

Advanced nuclear small modular reactors, 50–300 MW_e, and microreactors, up to 50 MW_e (which we collectively refer to as SMRs),[14–16] can be deployed modularly, i.e., one or multiple modules can be deployed at a given site. Due to this modularity,

CONTEXT & SCALE

Advanced nuclear small modular reactors (SMRs) are an emerging low-carbon technology that could help decarbonize industry, a hard-to-abate sector. We quantify the technoeconomic potential of SMRs in decarbonizing industrial heat by modeling the deployment and operation of five SMR designs at over 300 industrial facility processes in five US states. At 2021 US gas prices, we find that SMRs produce more expensive heat than natural gas at all industrial facilities. But at higher natural gas prices, SMRs can beat natural gas economically; a doubling of 2021 gas prices would unlock roughly 6 GW of SMR viability at industrial facilities. If SMRs simultaneously sell heat and electricity, a common strategy for industrial facilities, then SMRs can produce cheaper heat than natural gas for over 110 facility processes. SMR deployment would help decarbonize these facilities and potentially function as an early foothold for widespread SMR commercial deployment.

SMR deployment can be matched with industrial demand capacity requirements and can be mass manufactured and transported to the site, potentially reducing capital and deployment costs through economies of number and standardization.[14] Similar to existing co-generation plants at industrial facilities, SMRs can produce both direct thermal output and generate electricity.[13] Although no SMRs have been commercially deployed in the United States, several prospective reactors exist at various stages of design, regulatory approval, and construction.[17,18] Of these prospective reactors, four types are closest to deployment: integrated pressurized water reactors (iPWRs), integrated molten salt reactors (iMSRs), sodium-cooled fast reactors (SFRs), high-temperature gas reactors (HTGRs), and pebble bed high-temperature gas reactors (PBR-HTGRs).

Collectively, these SMR designs (iPWRs, iMSRs, SFRs, HTGRs, and PBR-HTGRs) have proposed outlet temperatures appropriate for decarbonizing industrial heat processes ranging up to approximately 850°C, considering heat transfer losses. Within this temperature range, SMRs would be able to serve petroleum refining, basic chemical, plastic, and ethyl alcohol facilities, which collectively emitted 180 million mt CO_2-equiv in 2014,[4] or 12% of total industrial emissions.[2] In addition to SMRs providing a technological solution to decarbonizing industrial heat processes, industrial applications of SMRs might provide a valuable niche for the early deployment of SMRs. By serving even a portion of the 650 GW of industrial primary energy usage in the United States,[19] SMRs could benefit from cost reductions enabled by economies of number and component-based learning.[20–22]

Several papers have considered industrial applications of SMRs at varying spatial scales, but the only studies that consider economic viability look at single remote facilities. McMillan et al.[4] found that, within the limits of technological feasibility, there is an opportunity to deploy 235 SMRs of 150 MW_{th} capacity, offsetting a third of the projected 2025 industrial thermal demands directly. Separately, they found over 800 SMR modules could be deployed within refineries, steel mills, and plastics industries, supplying both heat and hydrogen fuel for higher temperature processes through electrolysis. McMillan et al.[4] only considered technological feasibility and ignored the economic viability of SMR deployments. Similarly, without economic analysis, in the UK, Peakman and Merk[23] found low-temperature advanced reactors (such as light water reactors or iPWRs) could technologically meet a large amount of industrial heat demand, whereas only limited demand above 300°C was technologically feasible for higher temperature designs such as HTGRs and iMSRs. In contrast, a report by the Royal Society discussed a significant opportunity for nuclear across a number of applications above 300°C in the UK.[24] For applications less than 300°C, industrial participation and district heating have been studied and implemented globally for decades though in often isolated cases.[24,25] Other studies have combined the technical and economic aspects of this challenge when applied to a single, often remote facility or site. These studies have found mixed cost reductions from SMRs compared with existing diesel and natural gas use.[26–30] Friedmann et al.[5] found that there was only a limited opportunity for advanced nuclear in heavy industry, primarily due to the industries studied (e.g., steel and cement), which have temperature demands that exceed what SMRs can supply, and the uncertainty of costs. Outside of academic literature, Dow and X-energy have signed a letter of intent for the deployment of an HTGR at a Dow industrial facility, indicating economic potential for decarbonization.[31]

Despite the technical potential for SMRs to decarbonize a significant portion of US industrial heat processes, existing research does not quantify the technoeconomic potential for SMRs to serve industrial heat processes for more than a handful of

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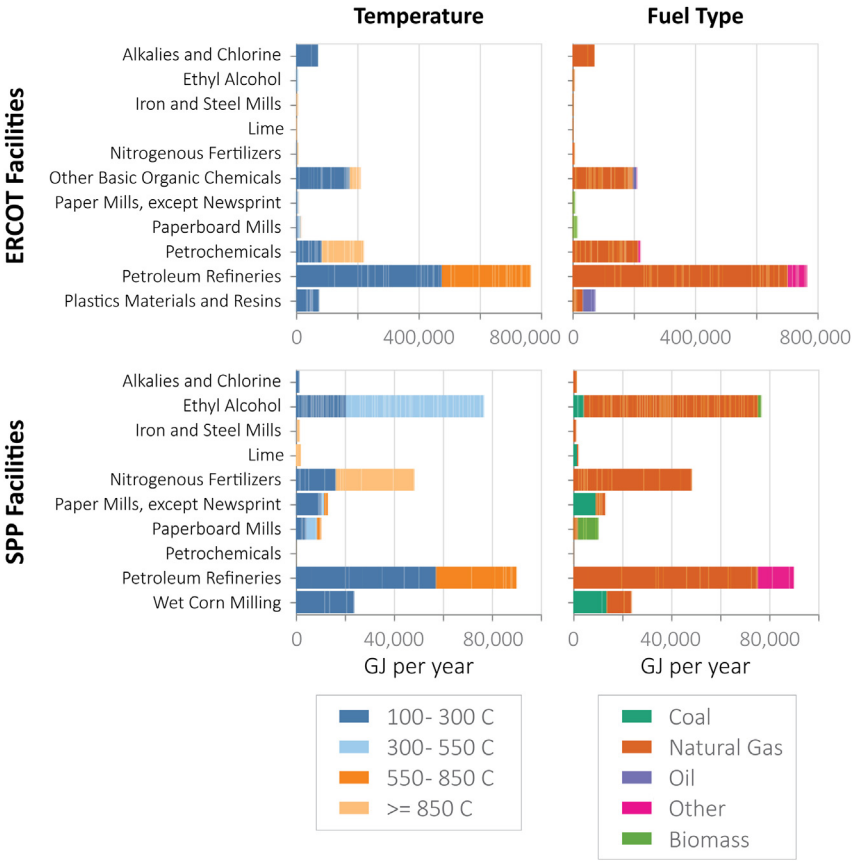


Figure 1. 2015 industrial thermal demand in ERCOT (Texas) and Southwestern Power Pool states by temperature bands (left column) and fuel types (right column)

Derived from McMillan and Ruth.[7] Each white line on the stacked bar indicates an individual facility process.

remote facilities. Quantifying technoeconomic potential indicates economically viable SMR applications, providing an understanding of likely rather than feasible deployments given the economically sensitive nature of industry. Furthermore, prior research has not considered SMR joint participation in industrial heat and electricity markets, a common strategy for existing industrial combined heat and power (CHP) facilities. This dual participation strategy could be a crucial driver of SMR economic viability in industrial applications, given the increasing electricity price variability with increasing renewables.[32]

We provide the first broad technoeconomic assessment of SMRs providing industrial heat by answering the following research questions: (1) Can SMRs achieve economic viability in decarbonizing industrial thermal demands? (2) How much does joint participation in industrial and wholesale power markets impact SMR economic viability? We designed a profit-maximizing mixed integer linear program (MILP) that optimizes, on a per-industrial-facility-process basis, the number and type of SMR modules chosen and the hourly operation of those modules. To capture heterogeneity across industrial facility processes and electricity markets, we analyzed over 300 grouped industrial facility processes at 180 industrial facilities in Texas, Oklahoma, Kansas, Nebraska, and South Dakota. These states span two electricity markets—the Electric Reliability Council of Texas (ERCOT) and the Southwestern Power Pool (SPP) (Figure 1). To capture heterogeneity in SMR designs, we analyze five

characteristic SMRs—an iPWR, an iMSR, and three HTGRs. For each reactor type at each industrial facility process, our MILP quantifies the annual net revenues of the chosen SMR when participating in industrial heat markets alone and in tandem with electricity markets. Our MILP enforces economically and technologically driven constraints on SMR deployment and operation, including that chosen SMRs at each industrial facility process must meet that thermal demand quantity (in MWh_{th}) and quality (in °C) on an hourly basis. By capturing thermal flows and efficiency penalties within SMRs, we optimize hourly thermal output, electricity output, and dump heat for each SMR. There are significant uncertainties associated with these nascent technologies and their integration with industrial facilities and processes. Although we cannot capture all unknowns, we evaluate the robustness of our results across interannual variability in electricity prices, SMR capital cost uncertainty, industrial demand variation, SMR reactor core flexibilities, the inclusion of thermal energy storage (TES), and the introduction of carbon pricing.

RESULTS

Economically viable deployment

To address our primary research question, we quantify the economic viability of SMRs to serve industrial loads within the SPP and ERCOT regions first without participation in wholesale power markets. Annualized costs of SMR deployments range from 7.66 to 855.07 million USD across facility processes in both regions. At 2021 US natural gas prices of roughly $4/MMBtu,[33] annualized revenues at these facility processes range from −4.76 to −534.05 million USD. No SMR types achieve positive net revenues at any of our 357 industrial facility processes that were studied. The average capacity factor of these SMRs is 0.9 (SD 0.2) and 0.8 (SD 0.3) for processes in SPP and ERCOT, respectively.

Given that no SMRs have positive net revenues at recent US natural gas prices, we identify breakeven natural gas prices at which the best suited SMR type would achieve positive net revenues for each group of industrial facility processes at each facility (see the SMR revenues from industrial heat section). We then order these facility processes by increasing breakeven prices to create economically viable "deployment stacks" by SMR type (Figure 2). We present the cumulative deployment stack across all SMR types as well as isolate how much of this is attributed to each SMR type, thereby providing insights into the deployments of SMRs in total and by design type. Limiting SMR deployment to one type per industrial facility rather than per each group of processes at each industrial facility does not qualitatively change our results (see supplemental information, section SMR type limited results, and Figures S17 and S18). We highlight results at three particular natural gas costs: $4/MMBtu, or recent US natural gas prices; $8/MMBtu, or roughly pre-fracking US natural gas (between 2007 and 2008 at $6.97/MMBtu and $8.86/MMBtu)[33]; and $16/MMBtu, the 2021 European Union (EU) natural gas price.[34]

In ERCOT, SMRs become economically viable at $6.56/MMBtu. At $8/MMBtu, 14% of modeled industrial demand can be decarbonized through the deployment of 27 PBR-HTGR modules (5.4 GW_{th}) at 10 industrial facility processes. At $16/MMBtu, 46% of modeled industrial demand can be decarbonized with 116 modules of mixed SMR types (21.1 GW_{th}) at 51 facility processes. In SPP, SMRs become economically viable at $7.00/MMBtu. At $8/MMBtu, 3% of modeled industrial demand can be decarbonized with a single PBR-HTGR module (0.2 GW_{th}), and at $16/MMBtu, 76% of modeled industrial demand can be decarbonized with 140 modules of mixed SMR types (7.55 GW_{th}) at 61 facility processes. Within SPP, although economically

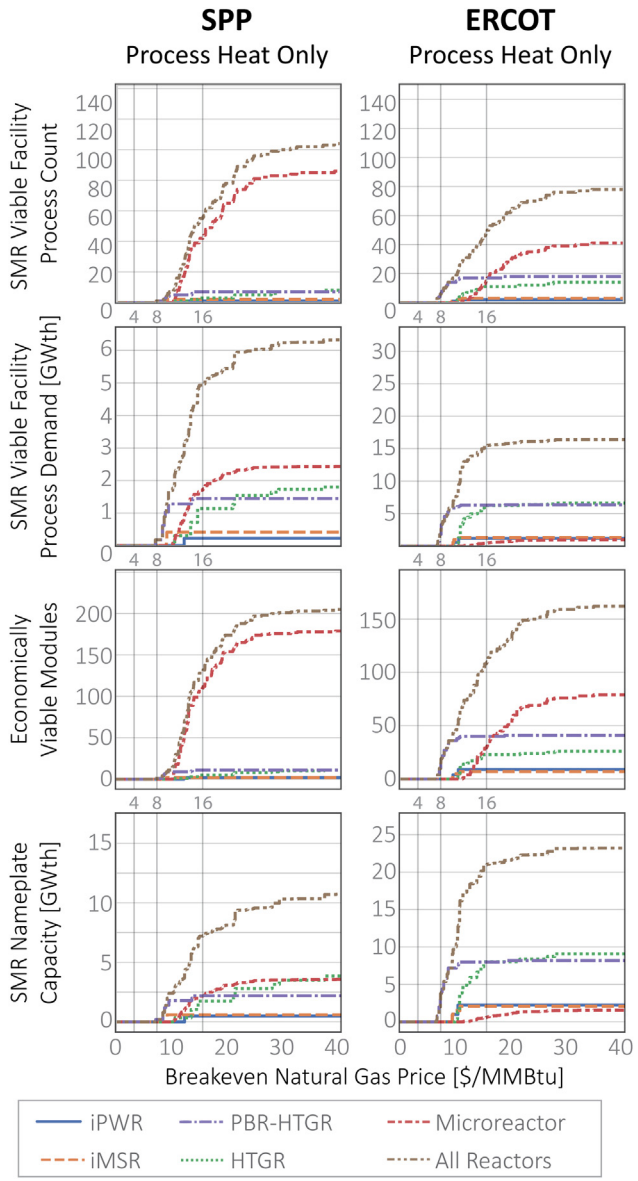


Figure 2. Economically viable deployment stacks of SMRs at industrial facility processes in SPP (left) and ERCOT (right) when providing process heat only across avoided natural gas costs

From top to bottom, we show the number of facility processes that could be economically served, the amount of facility process demand that could be served, the number of modules potentially deployed to serve the previous facility processes and demand, and the nameplate capacity of these cumulative modules across natural gas prices. Avoided natural gas costs of $4, $8, and $16/MMBtu are shown via dashed vertical lines.

viable SMRs are of mixed SMR types, 120 out of the 140 viable modules are microreactors.

Across both regions, the PBR-HTGR and microreactors are the most economically viable SMR types for serving industrial thermal loads due to their underlying costs, their capacity aligned with demand, and their outlet temperatures. Yet, if we only consider costs via the rough levelized cost of heat (LCOH) and ignore thermal quantity and quality requirements, the iPWR is the most economically viable option.

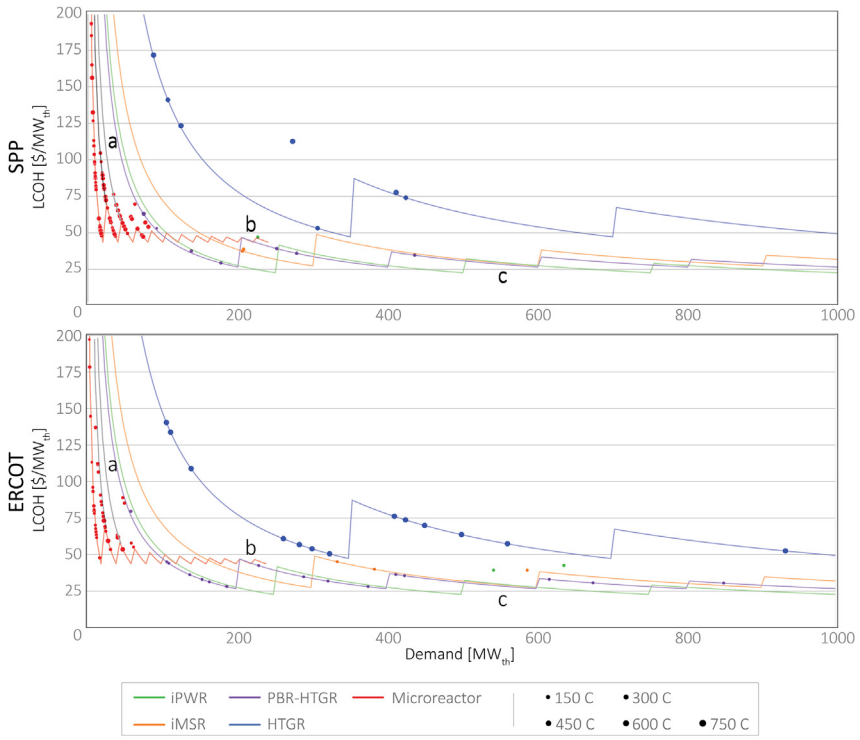


Figure 3. Comparison of the generalized levelized cost of heat (LCOH; lines) and our results (points)

The LCOH of each SMR type (shown with lines) decreases as the demand increases, with the exception of the upward steps when an additional module is introduced. Our results (shown with points of varying sizes for temperature demanded) closely follow the LCOH curves, and we typically choose the SMR with the lowest LCOH at its given demand and available temperature.

Specifically, a simplified LCOH calculation assuming a 100% capacity factor yields \$22.7/MW_{th} for the iPWR; PBR-HTGR, \$26.7/MW_{th}; iMSR, \$27.5/MW_{th}; microreactor, \$43.7/MW_{th}; HTGR, \$47.3/MW_{th}. But iPWRs are often not the most economically viable type of SMR for facility processes within our study system (Figure 1). For instance, at \$16/MMBtu natural gas prices, 11 iPWR modules are the most economical versus 245 modules of other SMR types. What then drives the selection of other SMR types? We identify two key factors that our model captures that a simple LCOH calculation does not: (1) thermal quantity and (2) thermal quality are required by industrial facility processes.

To understand the impact of the required thermal quantity, we extend our LCOH calculation across a range of constant demands and find that our results can be mostly explained by the matching of a specific facility's average demand (Figure 3). For each SMR type, there are declining LCOHs as demand and SMR utilization increase until a new module must be added, at which point there is a spike in LCOH (the solid color lines of Figure 3). We plot our results regarding the most economically viable SMR-type per-facility-process against these LCOH curves. In some cases, our results do not fall on the LCOH curves but instead along the implied extension of the curve when a higher number of modules are present (an example of this is shown with a gray line, noted as *a* in Figure 3). Because our industrial demand varies over time, peak demand exceeds average demand, requiring additional modules relative to our simple LCOH calculation. Below 100 MW_{th} of average demand, microreactors dominate the economically viable selections, even though they are the highest-cost

SMR type on a per-MW basis. This is due to the low utilization of larger, cheaper per-MW SMR types by these small facility processes, which results in higher LCOHs. Above 100 MW_{th}, the iPWR and PBR-HTGR become the lowest LCOH options, except for those between 250 and 300 MW_{th}, in which the iMSR is the lowest LCOH SMR type. These three SMR types all result in similar LCOHs of just over \$25/MWh_{th} beginning at 200 MW_{th} (*b* in Figure 3) and have less than \$2/MW_{th} in LCOH variation between 500 MW_{th} and 600 MW_{th}, (*c* in Figure 3). Due to their similar LCOHs, other operational factors can influence the model's decision of SMR type.

The second significant factor in the choice of the most economical SMR is which SMRs can provide the thermal quality required by a given industrial facility process. Only the HTGR can serve demands above 585°C. As a result, facility processes that demand heat above 585°C can only deploy the HTGR type of SMR, even though the HTGR has the highest LCOH among reactors (Figure 4). Due to their high LCOH, HTGRs only become economically viable at \$16/MMBtu (Figure 4). Thermal quality also limits iPWR deployment, which can only serve demands below 302°C (or 58% of total industrial thermal demand in SPP and ERCOT). In all scenarios and characteristic natural gas prices, the majority of economically viable demand (and, by proxy, the number of economically viable modules deployed) is below 585°C, and within this, the majority is below 300°C (Figure S15). Therefore, we find that the lower temperature facility processes provide more opportunity for SMR deployment. This finding is bolstered by our findings that, in a single SMR type to facility matching (see supplemental information, section SMR type limited results), the majority of SMRs are unaffected, whereas the HTGR opportunities are reduced.

Due to the quantity and quality of demanded heat, as described above, some industries are aligned with specific SMR types. For instance, ethyl alcohol facilities in SPP solely select microreactors at all natural gas prices due to their low thermal demand (see Table 2). Petroleum refineries initially select PBR-HTGRs at low natural gas costs, whereas at higher breakeven prices, more petroleum refineries adopt a more diverse set of SMRs (see supplemental information, section per-SMR type deployment by industry). For some industry types, such as petroleum refineries (Figure S4), the range of breakeven prices between the most economically viable and second best SMR type can be less than \$1/MMBtu with a full range across all technically feasible SMR types in the \$30/MMBtu range. These facility processes are those with demands above 200 MW_{th} and with highly similar iMSR, PBR-HTGR, and iPWR costs. Other industries, such as iron and steel mills (Figure S5) and basic organic chemical manufacturing (Figure S6), have much larger natural gas price differences between the most and second-most economically viable SMR types, and range across all technically feasible SMR types at over \$200/MMBtu (see Figures S4–S14). The SMR type decision at facility processes with large spreads in breakeven prices is more likely to be robust even considering uncertainty, whereas facility processes with smaller spreads are more likely to change SMR type decisions due to SMR and facility integration design uncertainties.

Industrial heat and power market participation

We answer our secondary research question by running the same model as above while allowing SMRs to provide process heat and participate in wholesale power markets. Relative to only participating in industrial heat, also participating in wholesale electricity markets increases the amount of economically viable industrial demand potentially met by SMRs and the deployable capacity of SMRs (Table 1). At 2021 US natural gas prices, 92 facility processes achieve positive net revenues when participating in heat and power markets, versus none when participating

Table 1. Economic viability of SMR deployment at representative natural gas prices

	ERCOT region					SPP region				
		SMR viable facility process demand (%)	Economically viable facility process count	Economically viable modules	SMR nameplate capacity (GW_{th})		SMR viable facility process demand (%)	Economically viable facility process count	Economically viable modules	SMR nameplate capacity (GW_{th})
$4/MMBtu	process heat only	0%	0	0	0	process heat only	0%	0	0	0
	w/ 2020 LMPs	0%	0	0	0	w/ 2021 LMPs	39%	113	113	33.9
Percent change		–	–	–	–	percent change	–	–	–	–
$8/MMBtu	process heat only	14%	10	27	5.4	process heat only	3%	1	1	0.2
	w/ 2020 LMPs	28%	16	55	11.0	w/ 2021 LMPs	69%	122	128	38.1
Percent change		108%	60%	104%	104%	percent change	2,524%	12,100%	12,700%	18,925%
$16/MMBtu	process heat only	46%	51	116	21.1	process heat only	76%	61	140	7.5
	w/ 2020 LMPs	64%	71	170	28.5	w/ 2021 LMPs	92%	127	136	40.9
Percent change		39%	39%	47%	35%	percent change	22%	108%	−3%	448%

only in heat markets. All of these positive net revenue SMRs are in SPP and account for 4 million mt CO_2-equiv across all facility processes; SMR annualized net revenues range from −524.8 to 10.8 million USD. The average capacity factor for these SMRs is 0.97 (SD 0.01) and 0.98 (SD 0.03) for facility processes in SPP and ERCOT, respectively. At $8/MMBtu, 84% and 10%, of SMR energy output is directed to wholesale power market sales totaling 126 TWh_e (16 GW_e peak) and 4.32 TWh_e (4.3 GW_e peak) sold annually within SPP and ERCOT, respectively (Table S3). If this quantity of electricity sales from SMRs were realized in SPP, electricity prices would decline, violating our price-taker assumption, so actual deployments would likely not reach these potential deployment levels.

To understand the sensitivity of SMR deployments to natural gas prices when participating in heat and power markets, we create SMR deployment stacks for increasing natural gas prices using the same method as in Figure 2. In ERCOT, using 2020 LMPs, SMRs would become economically viable at $6.56/MMBtu, the same as process heat only. At $8/MMBtu, there is a 108% increase in economical demand and a 104% increase in deployable SMR modules (up to 55 PBR-HTGRs at 11 GW_{th}) relative to process heat only. At $16/MMBtu, there is a 39% increase in economical demand and a 47% increase in modules (170 of mixed SMRs at 28.5 GW_{th}) relative to process heat only. Although participation in the wholesale power market significantly increases the number of economically viable SMR modules, it does not change which types of SMR modules are deployable at a given natural gas price, as PBR-HTGR and microreactors still dominate (Figure 5). In the SPP, 2021 LMPs, SMRs would become economically viable at $0/MMBtu natural gas prices, i.e., without industrial heat revenues, due to sufficient wholesale power market revenues. At $4/MMBtu, SMRs are economically viable for 39% of modeled thermal demands (113 facility processes) with 113 iMSR modules (33.9 GW_{th}). At $8/MMBtu, there is a 2,525% increase in economical demand and a 128-fold increase in modules (up to 128 modules at 38.1 GW_{th}) relative to process heat only. At $16/MMBtu, there is a 22% increase in economical demand and a 3% decrease in modules (136 of mixed SMRs at 40.9 GW_{th}) relative to process heat only.

Participating in both heat and power markets not only changes the economically viable SMR quantity relative to participating in only industrial markets in both markets but also significantly affects the type of SMRs deployable in SPP. When participating in heat and power markets, iMSRs are the dominant SMR type across natural gas prices, making up 100%, 94%, and 89% of deployable modules at $4, $8, and $16/MMBtu, respectively. When participating only in heat markets (Figure 2), PBR-HTGR and microreactors dominated deployments. Small, low- and medium-temperature facility processes choose the iMSR over the microreactor to maximize excess SMR capacity for participation in wholesale power markets.

Sensitivity analyses

We tested the robustness of our results against uncertainties in our technological and economic parameters for a reduced set of 50 randomly selected facility process demands across both regions (subset descriptive statistics in Table 2). Our results are robust due to constant demands, decreased flexibility, and the inclusion of TES. Additionally, our results are robust due to the inclusion of a heat transport loss across the majority of facility processes due to their excess capacity, though some individual facility processes experienced a modest increase in breakeven price (Figure 6). Our results are sensitive to the inclusion of carbon pricing and wholesale power market years (Figure S16). A US government carbon price of $51/mt CO_2-equiv[35] and Rennert et al.'s[36] value of $185/mt CO_2-equiv yielded increases in the economic

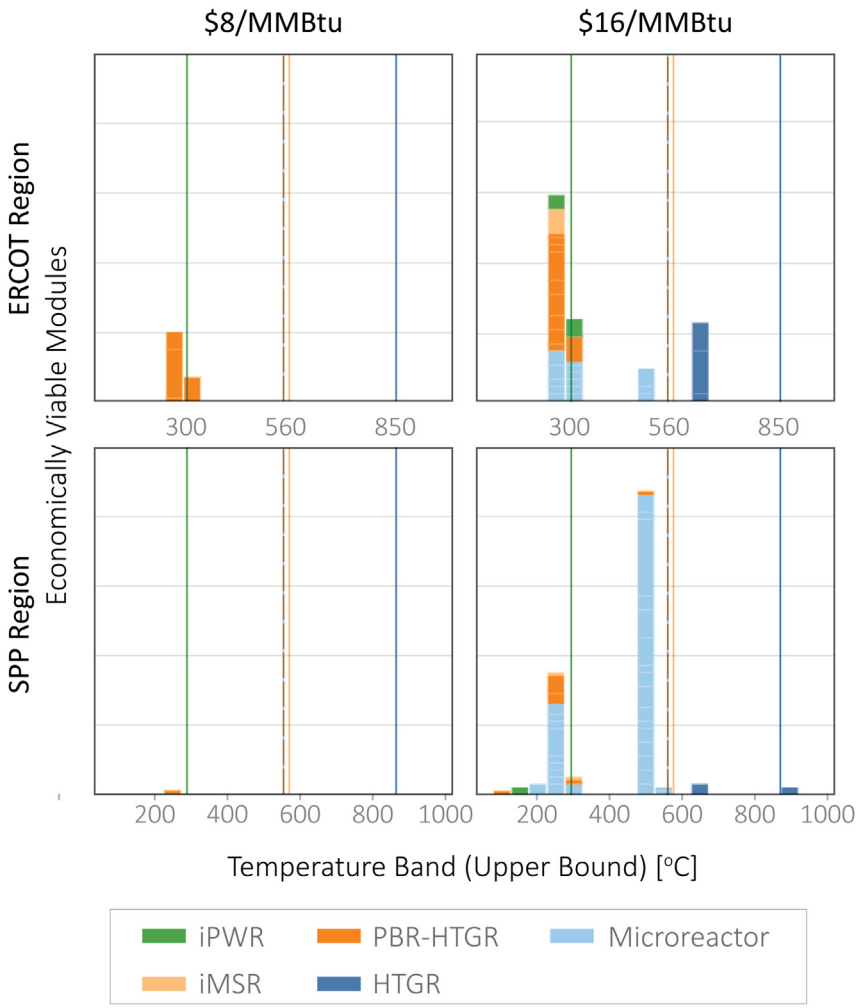


Figure 4. Number of economically viable modules of each SMR type across the required temperature range at representative natural gas prices

In each subgraph, the upper bound temperature of each SMR type is marked with a line of the same color as the SMR bars. In all cases, the highest temperatures require the highest natural gas prices to become economically viable with an SMR. Results with the inclusion of $4/MMBtu and wholesale market participation are available in Figure S7.

viability of SMRs in comparison to the existing natural gas. The US government value shifts the supply curve left by approximately $2.5/MMBtu, whereas Rennert et al. value shifts the curve by $9/MMBtu. The SMR type and general order of economical facility processes do not change with the carbon price. Using 2021 LMPs in either region, the highest price years, results in market parity for iMSR and PBR-HTGR SMRs and positive net revenues at 2021 US natural gas prices to varying degrees. Our results were insensitive to using other years of LMPs.

Given the significant uncertainty in future SMR capital costs, we evaluated 10%, 50%, and 100% increases in capital costs for all SMR types. We apply these increases to all SMR types uniformly, leaving type specific uncertainty for future study. At a 10% increase in capital costs, there is a minor reduction in SMR's economic viability (increase in breakeven price), though it is often similar to other sensitivities and the base case. At 50% and 100% increases, the deployment stacks are shifted rightward by roughly $4/MMBtu and $8/MMBtu, respectively. Given this rightward shift and

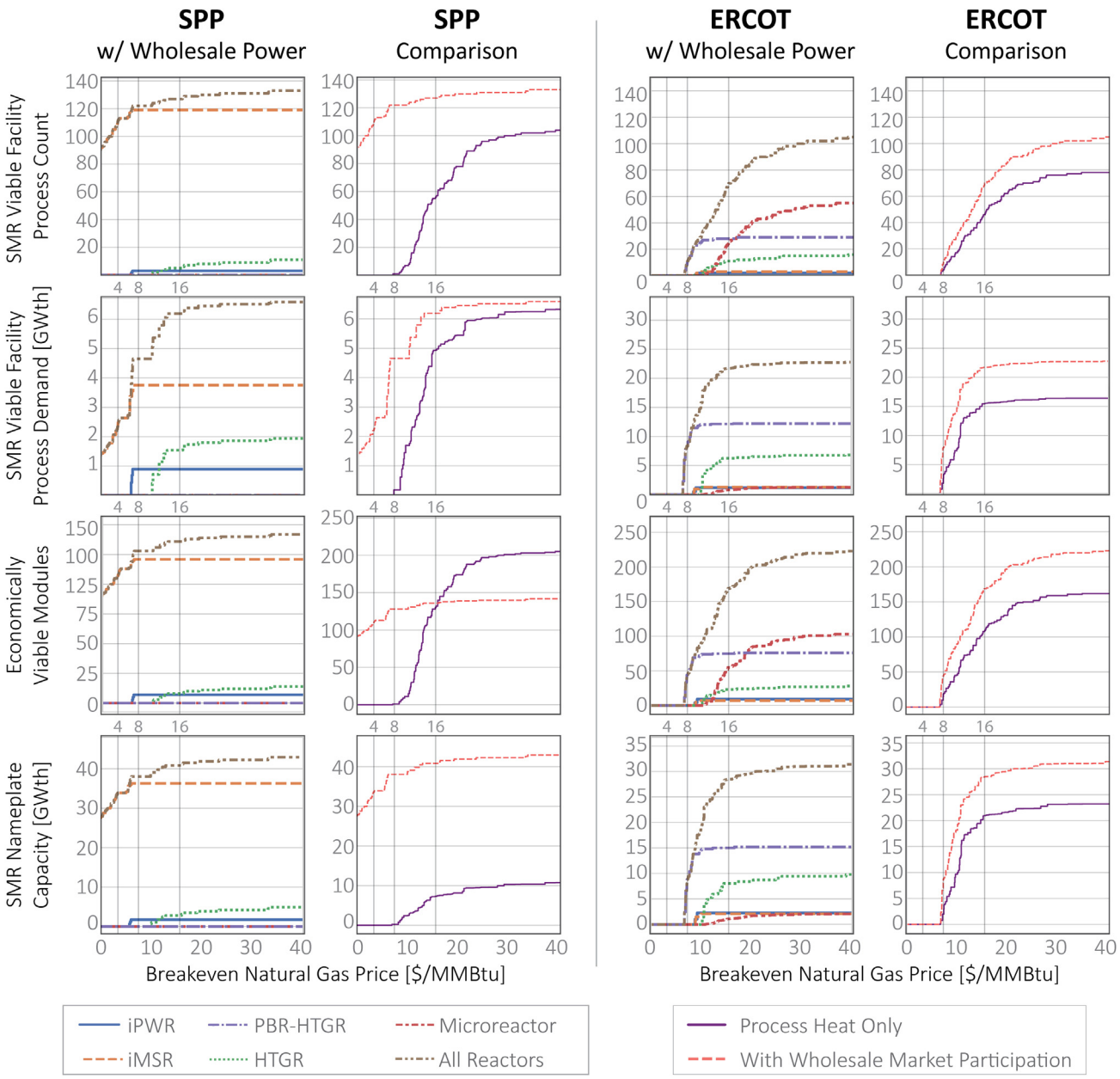


Figure 5. Economically viable deployment stacks for SPP (left two columns) and ERCOT (right two columns)

The left column for each region indicates deployment when participating in process heat and power markets, using the same results format as Figure 2. The right column for each region compares that deployment against that when only participating in process heat (from Figure 2).

the steep incline of the deployment stack, the economically viable facility process demand and the number of economically viable modules significantly decrease at each characteristic natural gas price we explore. For instance, in the SPP wholesale market participation case, SMRs no longer reach market parity at 50% and 100% capital cost increases.

DISCUSSION

We quantified the economic viability of SMRs for replacing existing fossil fuels in industrial high-heat processes when participating only in industrial heat or in industrial

Table 2. Descriptive statistics and characterization of natural gas fueled industrial processes for 2015 derived from McMillan and Ruth[7]

	SPP			ERCOT			Sensitivity set		
Number of facility processes	146			211			50		
Demand characteristics (MW_{th})									
Total	6,783			39,430			4,327		
Average	46.5			187.9			86.5		
Min/25th ptl/50th ptl/75th ptl/max	0/10/20/43/436			0/8/24/141/3420			1/10/19/40/934		
Emissions characteristics (million mt-CO_2/year)									
Total	11.2			66.6			7.3		
Per-industry characteristics									
Industry	SPP			ERCOT			Sensitivity set		
	Count	Total demand (MW_{th})	Temp req. (°C)	Count	Total demand (MW_{th})	Temp req. (°C)	Count	Total demand (MW_{th})	Temp req. (°C)
Iron and steel mills	4	50	180, 1,100	12	203	180, 1,100	2	34	180, 1,100
Petrochemicals	2	4	220, 875	39	6,800	220, 875	10	1,058	220, 875
Petroleum refineries	13	2,386	220, 600	41	22,270	220, 600	6	2,520	220, 600
Paperboard mills	10	55	150, 200, 275, 505, 800	5	19	150, 200, 275, 505, 800	3	6	150, 200, 275
Paper mills, except newsprint	14	125	150, 200, 275, 505, 800	5	22	150, 200, 275, 505, 800	3	32	150, 505, 800
Wet corn milling	2	319	50, 149	–	–	–	–	–	–
Other basic organic chemicals	–	–	–	59	6,256	260, 900	3	143	260, 900
Nitrogenous fertilizers	14	1,533	260, 850	2	220	260, 850	1	34	260
Alkalies and chlorine	1	43	177	2	2,277	177	–	–	–
Lime	2	18	120, 1,500	8	109	120, 1,500	–	–	–
Plastics materials and resins	–	–	–	30	1,046	260, 291	4	26	291
Ethyl alcohol	84	2,253	233, 450	8	211	233, 450	18	473	233, 450
Temperature distribution characteristics									
Temperature range	SPP			ERCOT			Sensitivity set		
	Count	Demand (MW_{th})		Count	Demand (MW_{th})		Count	Demand (MW_{th})	
≤ 300	74	2,959		118	24,696		29	1,538	
300–560	49	1,694		6	157		11	355	
560–850	19	2,067		26	8,989		4	1,316	
≥850	4	63		61	5,588		6	1,118	

The three sets described are facility processes in SPP, ERCOT, and the reduced, randomly selected 50 facility processes for our sensitivity analyses. "ptl" indicates percentiles, whereas "Temp req." indicates temperatures required at sub-processes within each industrial category.

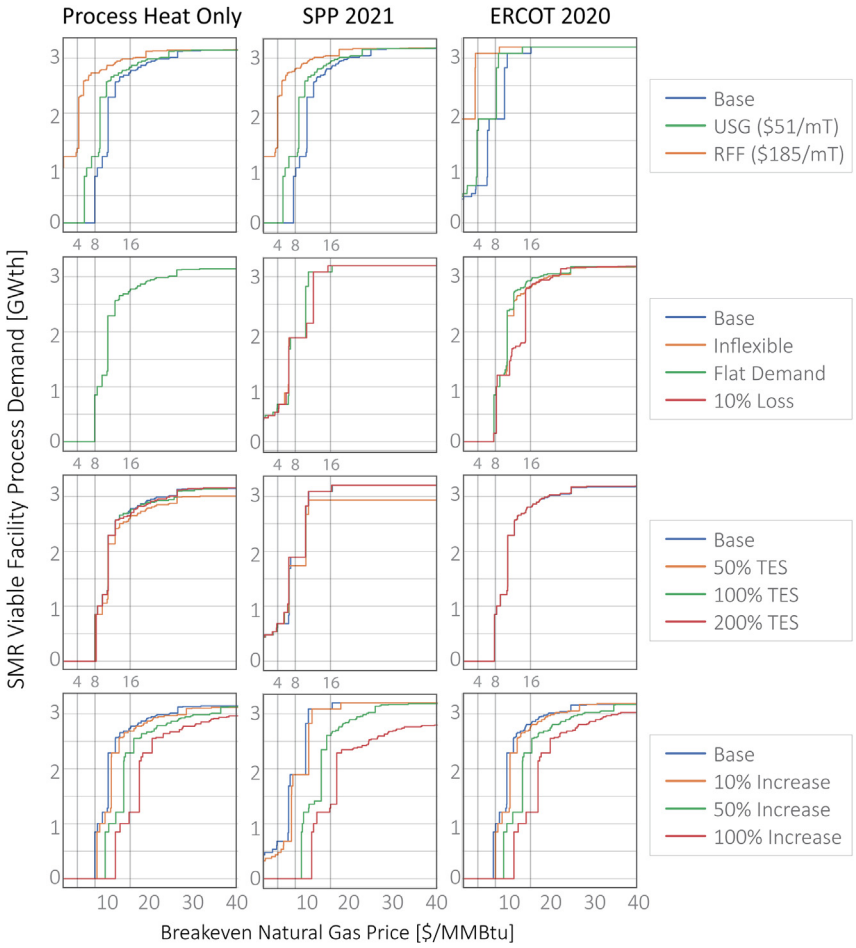


Figure 6. Economically viable facility process opportunities for SMRs across carbon price, SMR operational, TES, and capital cost sensitivities

Left column is process heat only, middle is 2021 SPP market conditions, and right is 2020 ERCOT. Each sensitivity is compared with our results, labeled "Base." In the first row, we include two carbon prices (Interagency Working Group on Social Cost of Greenhouse Gases, US Government[35] and Rennert et al.[36]). In the second row, we vary operational parameters (a constrained ≥90% capacity factor, a flat thermal demand, and a 10% thermal transport loss). In the third row, we vary the capacity of TES installed based upon the thermal demand (50%, 100%, and 200%). In the fourth row, we vary the capital cost parameter with a 10%, 50%, and 100% increase across all modules.

heat and wholesale power markets. We examined 357 industrial facility processes within the ERCOT and SPP wholesale power markets. We found that SMRs are not economically viable at 2021 US natural gas prices when only providing process heat to industrial facilities. At 2021 EU gas prices, 50% of the modeled thermal demands across both regions would be more economically served with SMRs than natural gas, resulting in over 250 economically viable SMR modules. With the inclusion of wholesale power market revenues, 5% of the modeled thermal demands would be more economically served by 113 SMR modules than existing natural gas at 2021 US natural gas prices, whereas 68% of thermal demands would be better served by over 300 SMR modules at EU natural gas prices. The PBR-HTGR and microreactor were the most widely economically viable for providing process heat only, with 51 and 157 modules, respectively, at $16/MMBtu across regions. In wholesale market participation scenarios, we found that two of the SMR types, the iMSR and PBR-HTGR, can reach market parity.

This research builds upon previous work that highlights the technical potential for SMRs in process heat and the limitations of SMRs in higher temperature industries.[4,5] In comparison with the 235 SMRs of 150 MW_{th} capacity, which McMillan et al.[4] found could decarbonize a third of industrial demands, we found a greater number of feasibly deployable modules and SMR capacity, even with the inclusion of economic viability and a reduced study region. This increase is due to the per-facility process analysis we performed, which only allowed an SMR to serve a single facility process demand, incentivizing the selection of smaller SMRs such as the microreactor. Friedmann et al.[5] only found limited viability for SMRs in industrial decarbonization in non-suited industries with temperature requirements exceeding 850°C. By focusing on process heat demands at 300°C–850°C that are better aligned with SMR outlet temperatures, we demonstrated that there is a technoeconomic potential for SMRs in industrial decarbonization. The potential is particularly substantial in petroleum refineries, chemical manufacturing, nitrogenous fertilizer production, and ethyl alcohol plants due to their capacity alignment with SMR designs and required process heat quality.

Past the economics and production uncertainty of SMRs, the question of waste remains. Any nuclear power source will inherently have both fuel wastes and wastes associated with decommissioned plant components.[37,38] The exact nature and quantity of SMR wastes and their associated processes are unknown, though it is likely that SMRs will produce more nuclear waste and fissile materials than their conventional counterparts, which can worsen nuclear waste concerns.[38] Others highlight the ability of SMRs to better control waste as the module and all waste within can be transported to a site specifically designed to process it.[37] Additionally, decommissioning costs will play a role in the investment decisions, though these are likely to be dominated by the other costs and revenues.[39,40] Deployment decisions for SMRs will need to wrestle with waste and other considerations in addition to economics, the focus of our analysis.

The challenge of decarbonizing industry to meet climate goals[1] is substantial, requiring a 65%–90% decrease in emissions globally. We found that when participating in heat and power markets, SMRs could economically decarbonize 38% of modeled industrial CO_2 emissions in SPP at 2021 natural gas prices. Further decarbonization via SMRs would require policy support. At a $51/ton carbon tax, for instance, SMRs could decarbonize 66% of industrial CO_2 emissions in SPP. Other industrial deep decarbonization options, such as green hydrogen or biomass, are economically similar to or less viable than SMRs, with breakeven costs above $30/MMBtu and $10/MMBtu, respectively.[5] Thus, based on our analysis, SMRs could provide an initial industrial decarbonization technology for SPP. The US Inflation Reduction Act (IRA) provides two further policy supports—a $25/$MWh_e$ production tax credit for electricity supplied to the grid for their first 10 years or a 30% investment tax credit that could further improve the technoeconomic potential of SMRs in decarbonizing industrial heat under specific employment guidelines.[41]

Our two study regions combined contain 30% of the national industrial thermal demand from Environmental Protection Agency's Greenhouse Gas Reporting Program (EPA GHGRP) facilities.[7] To estimate the nationwide economic viability of SMRs at industrial facilities, we scale our results in reference to regional demand to other power market regions, namely Michigan, Minnesota, Wisconsin, Iowa, Illinois, Indiana, Mississippi, Missouri, Louisiana, Arkansas, and North Dakota (MISO); Ohio, Pennsylvania, Kentucky, Virginia, New Jersey, West Virginia, Maryland, and Delaware (PJM); and California only (CAISO), assuming similar power markets and industrial facilities.

There is a cumulative 92.6 GW_{th} reported industrial demand across these three regions,[7] which, proportionally, at EU natural gas prices and participating in heat and power markets, could be viably served by 611 SMR modules. Across all these markets and our studied regions, that could result in up to 101 million mt CO_2-equiv, or 7%, of total US industrial emissions avoided annually. This rough estimate underlines the potential for SMRs in industrial decarbonization and widespread usage.

When focusing on facility process characteristics that define economic viability, we found three defining factors: (1) thermal demand quantity matching, (2) thermal quality requirements, and (3) wholesale power market, when available. Without wholesale power market participation, thermal demand quantity matching drives utilization of installed modules, which in turn drives net revenues. In particular, industrial facility processes larger than 100MW_{th} offer the most economically viable SMR deployment opportunities due to their alignment with the lowest-cost SMRs (iMSR, iPWR, and PBR-HTGR), with few exceptions for well-matched microreactors. These large EPA GHGRP facilities account for less than 1% of US industrial facilities but for roughly 25% of total US industrial emissions, representing a valuable area to focus decarbonization measures. Additionally, the scale of SMRs could lend themselves to industrial clusters of smaller industrial facility processes with different owners and different temperature needs that co-located to take advantage of these resources.

Our study can serve as the basis for future detailed studies of SMR deployment and integration with specific industrial facilities. Although we account for gross thermal demands and temperatures required by each studied facility process, we lack data on the spatial layout and infrastructure of the facilities. These factors will likely impact SMR deployment and integration and are best studied for specific location case studies with an industrial partner. Through this type of case study, each subprocess, rather than bundled demand, could be treated as an area of optimization with associated losses and heat quality requirements. Our analysis of the broad opportunities for SMRs suggests such case studies should initially focus on basic organic chemicals, petrochemicals, and refineries in ERCOT with PBR-HTGR developers and ethyl alcohol facilities in SPP with microreactors, as these deployments are the most economically viable in our analysis when providing only heat (see Figures S4–S14).

Although we have tested our results against some uncertainties, our model is limited in multiple ways, including (1) limitations of historical market data and (2) SMR cost assumptions. We use historical natural gas and electricity prices, but both markets are subject to significant uncertainty due to a decarbonizing power system and geopolitical factors. Future work could include a more robust analysis of LMPs under a range of renewable penetrations and power system assumptions or power purchase agreements for delivered heat and/or electricity. With respect to SMR capital costs, we found our results were robust to minor increases in capital costs but are impacted by increases of 50% and above across all SMR types. Given that SMRs have yet to be commercially deployed, SMR capital cost estimates remain speculative to some degree. Moreover, based on their development stage, designs, prefabrication strategy, and other factors, different SMR designs likely have different levels of cost uncertainty. As SMRs and their capital cost estimates mature, future work can provide more accurate estimates of SMR economic viability. To that end, SMRs are designed to benefit from learning curves and economies of number,[20–22] but initial installations must occur to start this process. Using the economic viability of SMRs in this work, a staged, multi-year analysis of installations and SMR learning could evaluate how impactful economies of number will be on overall SMR economic viability

in process heat supply, wholesale power market participation, and combined applications.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the lead contact, Max Vanatta (mvanatta@umich.edu).

Materials availability

This study did not generate new unique materials.

Data and code availability

Full methods and model formulation are available in supplemental information sections, model formulation and thermal energy storage (TES) constraints and data. Detailed code and data are available at https://doi.org/10.5281/zenodo.7752736. For detailed instructions on running model, please contact the corresponding author.

SMR viable deployment and operations model

To assess the technoeconomic value of SMRs in industrial heat applications, we quantify the annualized net revenues of SMRs on a per-industrial-facility-process basis for two market scenarios: (1) when providing process heat and (2) when providing process heat and participating in wholesale electricity markets. We design a profit-maximizing MILP that optimizes, on a per-industrial-facility-process basis, the number and type of SMR modules chosen and the hourly operation of those modules, i.e., their on/off state, thermal generation, and electricity generation. The model chooses from five characteristic SMRs (see the SMR design parameters section). Our model's objective function is to maximize annualized net revenues (Equation 1):

$$maximize: \pi - c \quad \text{(Equation 1)}$$

where π = annual revenues ($) (Equation 2) and c = costs ($) (Equation 3), which are defined as:

$$\pi = \sum_{g,n,t} e_{g,n,t} \times LMP_t \quad \text{(Equation 2)}$$

$$c = \sum_{g,n,t} \left(\left(SC_g \times v_{g,n,t} \right) + \left(VC_g \times r_{g,n,t} \right) \right) + \sum_{g,n} \left(Cap_g \times m_{g,n} \times \left(OCC_g \times CRF + FC_g \right) \right) \times D \quad \text{(Equation 3)}$$

where t, g, and n index hourly timestep, SMR type, and module within maximum module count, respectively; e, hourly electricity generation (MWh_e); LMP, locational marginal price of electricity ($/MWh); SC, startup cost ($/startup); v, turn on binary variable; r, total reactor output (sum of heat and power) (MWh_t); VC, variable cost composed of variable operations and maintenance (O&M) and fuel costs ($/$MWh_t$); Cap, nameplate capacity (MW_t); FC, fixed costs ($/$MW_t$/year); OCC, overnight capital cost ($/$MW_t$); m, binary indicator of built modules; CRF, capital recovery factor (see supplemental information); and D, scaling factor to account for length of optimization (explained below).

By not including industrial heat revenues in our objective function, we can quantify the sensitivity of SMR economic viability to various natural gas prices and identify breakeven

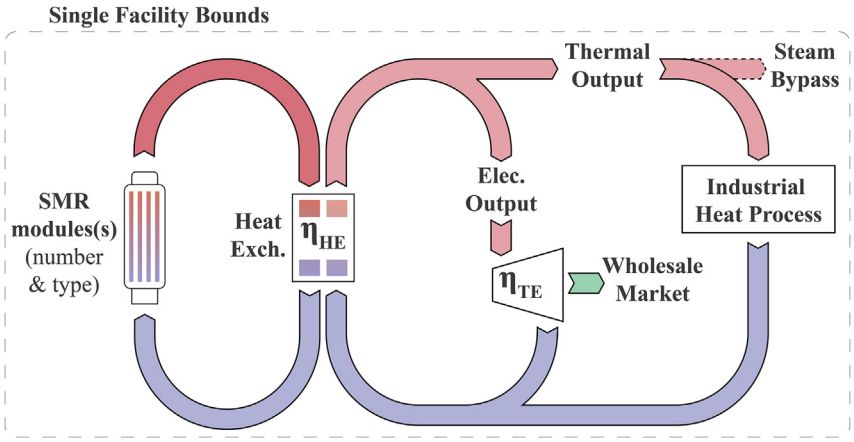


Figure 7. Diagram of a single SMR module system

Multiple module operations would appear the same, but all industrial heat process outputs would be pooled between modules.

natural gas prices at which SMRs become economically viable (see SMR revenues from industrial heat). In calculating electricity revenues, we assume that the wholesale price for electricity is unaffected by SMR operations (see wholesale electricity market data).

Our model constraints enforce economically and technologically driven limits on SMR deployment and operation. For each facility process, the model must deploy and operate SMRs to meet thermal demand quantity (in MWh_{th}) and quality (in °C) on an hourly basis. In joint wholesale market participation scenarios, the model decides how much of the remaining capacity above the required thermal demands to commit to electricity generation. By capturing thermal flows and efficiency penalties within SMRs, we optimize hourly thermal output, electricity output, and dump heat for each SMR (Figure 7). We also constrain SMR reactor and electricity generation operations through engineering-driven unit commitment, ramping, and minimum load constraints (see SMR design parameters). We constrain our model to deploy only one of five potential SMR types at a given industrial facility process (see SMR design parameters). The number of built SMR modules is bound between zero and a maximum value, where the maximum value equals the facility process's peak heat demand divided by each type of SMR module's thermal output capacity plus a ten percent surplus rounded up to the nearest larger integer. This maximum value reduces the number of model solutions where SMRs are optimized as a merchant generator rather than as an industrial heat provider. We test the sensitivity of our results to uncertain parameterizations of key constraints (see the sensitivity analysis description and Table 5). The heat demand dataset we use relies on observed fuel and heat usage, so we indirectly account for existing system loss due to plant operation within our demand calculations. We do not consider the design, losses, or costs associated with the heat transfer and distribution system beyond what is already captured in the existing heat demand calculations. A more rigorous siting analysis per-facility process, including minimum setbacks and context, is left for later investigation, though we test the sensitivity of this additional loss (see Table 5).

The full model formulation is available in the supplement information. We formulate the model in Pyomo[42] and solve it using IBM CPLEX.[43]

Due to computational constraints, we cannot run our model at hourly resolution for the entire year. Instead, we run our model twice for each industrial facility process. In the first run, we approximate annual operations to optimize viable SMR deployment (type

and number of modules) at each industrial facility process. To approximate annual operations, we run the model every sixth hour of the year (i.e., for 1,460 h). Because we run for only 1,460 h, we scale down annualized capital costs (term *D* in *Eq 3*). In the second run, we fix the SMR choices (type and number of modules) from the first run for each industrial facility process, then optimize operations for each hour of the year (i.e., for 8,760 h). To approximate participation in day-ahead electricity markets, we run our model for a rolling 48-h optimization horizon consisting of a 24-h optimization window and a 24-h look-ahead period. We assume perfect foresight of LMPs during this horizon, a common assumption in price-taker models.[44,45] The final period of each optimization window initializes the next 48-h optimization horizon. For the last day of the year, we only run a 24-h optimization window.

Industrial facility process thermal demand

We obtain industrial thermal demand data for 200 facilities in the ERCOT (Texas) and SPP (Oklahoma, Kansas, Nebraska, and South Dakota) regions, spanning 12 industries, including petroleum refineries, ethyl alcohol, and basic organic chemicals. These facilities contain 423 heat processes requiring a total of 52.7 GWh_{th}.[7] The thermal demand data are provided on a per-facility and per-process basis by the US National Renewable Energy Laboratory (NREL).[7] The NREL data correspond to facilities and processes registered with the US EPA GHGRP in 2015, the most recent year of data available in the published dataset. To be registered under the GHGRP, a facility must directly emit more than or equal to 25,000 mt CO_2-equiv per year.[7] This dataset also includes fuel usage, location, and required temperature. Although this constrains our study to 1% of national facilities,[4] due to the costs and other challenges associated with SMR deployment, it is unlikely that small facilities would have the resources to operate standalone systems. Furthermore, EPA GHGRP facilities account for 25% of annual industrial emissions.[4] As detailed in the results, facility processes smaller than 1 MW_{th} are unlikely to provide economically viable deployment opportunities for SMRs. As a result, missing these facility processes likely does not affect our insights regarding SMR technoeconomic viability in industrial heat applications.

The NREL dataset provides process-level energy demand within each facility, which can result in many similar yet distinct demands at a given facility. Individual processes might be too small to be economically viable for SMR deployment, but aggregated process demands at the facility level might be economically served by an SMR. We filter and combine process-level thermal demand data in several ways. First, given that 88% of this thermal demand is provided by natural gas,[7] we limit our analysis to heat processes using natural gas (see Figure 1). Second, we combine individual processes of the same temperature into a single temperature demand and treat thermal demands larger than 500 MW_{th} (large enough for multiple modules of each SMR type) as a unique deployment opportunity. We perform this operation because facilities will likely maximize the utilization of capital-intensive SMRs by using them for as many processes as possible with similar heat quality. There are 146 resulting batched process demands, henceforth referred to as facility processes, at 69 facilities in SPP (63 have multiple processes) and 211 facility processes at 111 facilities in ERCOT (94 have multiple processes) (described in Table 2). Of these, 245 facility processes across regions are within the 0–50 MW_{th} range (see Figure S1).

In the batching process, we do not include the transportation losses associated with the geographic distance between processes at a single facility. This assumption may underestimate the required SMR output to meet thermal demands. This assumption of multiple batched processes at a single site may result in smaller SMR types being preferred over larger SMRs due to full facility demand being higher than component processes.

Table 3. SMR parameters

Reactor	iPWR	HTGR	PBR-HTGR	iMSR	Microreactor
Outlet temp (°C)	302[53]	950[50]	750[54]	700[55]	630[56]
Steam temp (°C)	302	852	563[54]	585[57]	565[56]
Electric power (MW_e)	77[58]	164[50]	80[54]	141[59]	6.7[27]
Thermal power (MW_t)	250[58]	350[50]	200[54]	300[59]	20[27]
MSL (%)[a]	20[60]	20	40[54]	20	20
MSTL (%)[b]	10[53]	10	10	10	10
MDT (h)	20	20	20	20	20
Ramp rate (%/h)	40[61]	600[62]	240[54]	60[63]	600[62]
Overnight capital cost ($/$kW_e$)	5,535[64]	7,500[17]	4,569[65]	4,091[59]	10,902[27]
Fixed O&M cost ($/$kW_e$-year)	115[66]	164[50]	100[67]	85[55,68]	264[27]
Variable O&M ($/$MWh_e$)	0.75[66]	0[c]	0[c]	0.5[55,68]	0[27 c]
Fuel cost ($/$MWh_e$)	9	13[50]	13[50]	11[59]	0[27 d]
Startup fixed cost ($/$MW_e$/startup)	500	500	500	500	500

Outlet temperature indicates the high temperature of the coolant loop, which is then transferred to the steam working fluid with a thermal transfer penalty. All minimum down time (MDT) values are assumed to be 20 h due to a dearth of more specific values. Minimum stable load (MSL) is the lowest reactor output level, whereas minimum stable turbine load (MSTL) power is possible in the electricity-specific loop. The MSTL can be lower than the MSL as an SMR can produce both process heat and electricity simultaneously. All 2020 $ and FOAK costs.

[a]20% of nameplate[60,69]

[b]10% of nameplate[53]

[c]Considered 0 due to all O&M allocated to fixed costs.

[d]These microreactors have no refuel, and their lifetimes are bound by initial fuel. As such, reported fuel costs are $0.

We leave specific transition studies, considering different batching methods and facility design specifics, for future work.

To disaggregate facility-process-specific thermal demand from an annual to hourly basis, we use industry-specific hourly thermal demand profiles where available and otherwise assume constant thermal demand. McMillan[46] derived industry-specific thermal demand profiles for some of our studied industries using the Census Quarterly Survey of Plant Capacity Utilization,[47] Industrial Assessment Center (IAC) Database,[48] and EPA Air Markets Program Data (AMPD).[49] They used seasonal weekly operating hours from the Census Quarterly Survey data to create a daily and weekly on/off schedule, assuming nights and weekends accounted for "off" hours. Large industrial facility processes with high heat processes do not often shut down fully, so during "off"-time periods, McMillan[46] assumed a minimum demand of 25% of the capacity based on EPA AMPD data. The result of this process is a normalized system capacity usage of 0 to 1 (see Figure S2). When available, we apply these variable hourly thermal demands, which are provided by day of the week for each month, to generate a full annual hourly timeseries of normalized capacities for each calendar year. We then scale the hourly demand value for each process such that the annual thermal demand equals that from McMillan and Ruth.[7] We test the robustness of our results to variations in heat load curves in the sensitivity analysis using a flat demand for all hours (see Table 5).

SMR design parameters

Through literature review of white and gray literature and discussion with SMR developers, we parameterize five characteristic SMR types that are closest to deployment and cover a wide range of outlet temperatures: one iPWR, one iMSR, and three HTGRs (see Table 3).[13] We opt to not include SFRs, such as TerraPower's Natrium, at this time due to their close integration with TES systems,[13] leaving them for future

research. iPWRs are currently in the development process, with NuScale and INL anticipating a 2027 completion date for grid connection.[18] These reactors use light water as the coolant for the core, which then transfers energy to an outside working fluid.[13] iPWRs have reactor outlet temperatures of roughly 300°C.[13] iMSRs use molten fluoride salts as the coolant, transferring energy to a secondary working fluid.[13] iMSRs have reactor outlet temperatures between 540°C and 800°C.[5,13] HTGRs use a gas such as helium as the coolant that transfers energy to a secondary working fluid.[13] X-energy, a company producing HTGRs, is set to complete construction of a facility by 2027 as part of the US Department of Energy's Advanced Reactor Demonstration Program.[17] HTGRs have prospective reactor outlet temperatures of 650°C–950°C[5,13,50] though the highest temperatures will require material development. To provide generalized insights, these characteristic SMRs represent a generic reactor of each type rather than a specific company's product, although we gather company-specific data where available through interviews and public materials. SMR parameters include operational parameters (e.g., minimum stable load, outlet temperature, thermal efficiency) and economic parameters (e.g., capital cost, fixed O&M, variable costs).

Given that no SMRs have been commercially deployed, our economic parameters are subject to significant uncertainty related to direct capital and operational costs, build schedules, and other factors. For this work, we use FOAK (first-of-a-kind) costs for all SMRs with the knowledge that the NOAK (nth-of-a-kind) costs that may be leveraged for later industrial heat supply could be lower. Additionally, we assume an overnight capital cost (OCC) (i.e., overnight construction of the SMR), which neglects the interest accrued over the build timeline. The build schedules for existing nuclear power plants are often uncertain due to construction contingencies, which increase costs when they arise. SMRs are intended to mitigate, but will likely not eliminate, construction-related uncertainty.[51] Further, the OCC of these SMR types includes the SMR vessel itself, the structure, building, and balance of plant necessary to operate as a power plant (e.g., turbines and electrical equipment). This will increase the cost compared with thermal generation alone, but the itemized distribution of the costs across component types is highly dependent on design, which falls outside of our scope. Similarly, our assumed costs are also not module number specific in that they do not scale based upon the number of modules at a single site, which could share certain components and reduce the per-kW cost for larger packs of modules. The result of this assumption in our model will be moderately favorable breakeven prices for single module deployment versus larger module counts, though by how much is outside the scope of this study. We neglect SMR decommissioning costs due to a lack of literature estimating those costs[40] and the likely marginal impact of their present value compared with other costs and revenues.[39,52] As these parameters are subject to significant uncertainty, we perform a sensitivity analysis (see Table 5), applying a 10%, 50%, and 100% OCC increase, but given the immaturity of the technology, we do not evaluate an OCC reduction case.

SMR revenues from industrial heat

As detailed above, we choose not to include industrial heat revenues in our objective function, so we can instead quantify SMR economic viability across a range of avoided natural gas prices. In displacing the use of natural gas for process heat, we assume SMRs will be compensated by each industrial facility process at a marginally lower cost than the one each facility process would have paid for the displaced natural gas, which can be expressed as:

$$SMR\ Industrial\ Revenues = \frac{Demand}{AHF} \times P \qquad \text{(Equation 4)}$$

Table 4. DAM electricity wholesale market descriptive statistics, reported on an hourly basis

		Electricity price ($/MWh)						
Market	Year	Mean	Median	SD	Min.	Max.	1st quartile	3rd quartile
ERCOT	2018	33.07	24.49	75.52	4.20	2,061.56	19.65	32.48
	2019	37.62	20.51	164.15	2.91	5,008.13	16.65	27.18
	2020	**20.23**	**16.57**	**38.53**	**−24.46**	**1,387.64**	**13.05**	**19.49**
	2021	145.62	27.58	891.63	−0.25	8,995.26	20.31	39.57
SPP	2020	17.79	17.20	9.53	−13.19	77.75	12.98	22.11
	2021	**67.65**	**25.84**	**327.17**	**−13.83**	**4,460.06**	**17.29**	**39.88**

where *Demand*, annual heat demand (MMBtu); *AHF*, available heat fraction; and *P*, natural gas price ($/MMBtu).

To quantify displaced natural gas on a facility-process-level basis, we convert each annual quantity of thermal demand (from the prior section) to a quantity of natural gas that must be combusted in MMBtu. The efficiency of the heat process varies based on process temperature, which we capture using a temperature-dependent available heat fraction (*AHF*) coefficient assuming 10% excess air[70] (Equation 5):

$$AHF = -0.00038T + -0.00000T^2 + 0.90556 \quad \text{(Equation 5)}$$

where *T*, process temperature (°C). We assume that fuel costs dominate variable process heat costs, so do not include any O&M costs. By quantifying the SMR industrial revenues and comparing them with the optimized operation costs of the SMRs, we can derive a net revenue for an SMR-served facility process that includes avoided natural gas costs. This allows us to economically compare existing fuel usage to an SMR retrofit.

We use this same process to find an avoided natural gas breakeven price, henceforth called the breakeven price, at which the SMR is a more economically viable source of industrial heat than existing natural gas annually. We find this value by dividing the post-optimization net revenue of the SMR system by the demanded natural gas quantity, resulting in a $/MMBtu value. This value allows us to understand how many facility processes, how much demand, and how many SMRs are economically viable at industrial facility processes given a selected natural gas price.

Wholesale electricity market data

In our second market participation scenario, SMRs can provide process heat to industrial facility processes and sell electricity into day-ahead wholesale electricity markets. To quantify revenues from electricity market participation, we use hourly day-ahead market (DAM) electricity prices from the market (ERCOT or SPP) that each facility is located in. Because we don't have electricity nodes corresponding to each industrial facility, we use hub average prices rather than nodal LMPs. In ERCOT, we use "HB_HUBAVG."[71] In SPP, no hub average is supplied, so we instead average all node LMPs on an hourly basis.[72] Using hub average prices could misestimate SMR revenues from electricity market participation, but attempting to match industrial facilities to nodes would introduce significant uncertainty into our analysis.

We operate our model assuming the industrial facility is a price-taker, meaning it accepts the given hourly wholesale electricity price without affecting it. This price-taker approach is computationally light, allowing us to model SMR deployment and operation in detail at hundreds of individual industrial facility processes. Price-taker approaches are also common in technology assessment and strongly justified when

Table 5. Sensitivities descriptions and rationales

Sensitivity	Description	Rationale
Wholesale market year	we evaluate our model across 6 years of wholesale power market data. We use ERCOT data from 2018 to 2021 and SPP data from 2020 to 2021 (see main text Table 4).	wholesale power markets have the potential to provide a valuable revenue source, but these markets are uncertain and variable.
Capital cost uncertainty	we increase capital costs by 10%, 50%, and 100%.	as no SMRs have been completed, uncertainty surrounds SMR capital costs.
Industrial demand variability	we test constant industrial demand against hourly variable demand.	incomplete information on the hourly demand of industrial process heat at all of our studied facility processes exists.
Reactor core flexibility	we increase the MSL to 90% of capacity.	SMR designs remain uncertain and could be less flexible than we estimate.
Thermal energy storage	we quantify the net revenue impacts of including a simple TES system into the model sized in proportion to the facility process demand at 0, 50, 100, and 200% of average demand.	TES can be leveraged to increase flexibility while also satisfying inflexible thermal demands, allowing greater market revenues.
Carbon pricing	we include two social costs of carbon: \$51/mt-CO_2 and \$185/mt-CO_2.	carbon pricing to incentivize decarbonization could improve the economic viability of SMRs supplying industrial process heat.
Working fluid transport/ distribution energy loss	we include a 10% energy reduction[74] in the supplied thermal energy to account for increased transport distances.	after heat is produced by the SMR, the working fluid carrying the heat to the industrial process must pass through a transport/distribution system that will have losses. Some losses are already included in the existing dataset, but increased transport distances for nuclear setbacks could increase the losses.

analyzing the participation of small generators in wholesale markets.[73] Substantial deployments of SMRs into these wholesale markets would break the price-taker assumption and overstate the economic viability of participating SMRs. An alternative approach would be to embed industrial facility processes and SMR deployments and operations within a power system planning or operational model. But these models lack computational tractability, which would require simplifications to our modeling of SMRs and industrial facility processes and often result in poor reproduction of historic prices. Future work should include some kind of feedback between the industrial facility process deployment of SMRs and the changing power system to better understand how a price-maker's (rather than our price-taker's) assumption would impact results.

For our base analysis, we use 2020 prices from ERCOT[71] and 2021 prices from SPP.[72] These years capture relatively low (2020 ERCOT) and moderate to high (2021 SPP) wholesale prices (see Table 4). These market years are chosen not to be directly comparable but instead to highlight the differences in economic viability due to different power market conditions. We test the robustness of our results across additional price years from both markets via sensitivity analysis (see Table 5). We exclude 2021 ERCOT prices from our main analysis (but include it via sensitivity analysis) due to extremely high prices driven by Winter Storm Uri, which are already represented in the 2021 SPP prices.

Scenarios and sensitivities

To answer our research questions, we use scenario analysis to modify the available sources of revenues for the facility-based SMRs. We run two scenarios: (1) SMRs only serving industrial process heat demands, and (2) SMRs jointly serving industrial process heat demands and participating in wholesale electricity markets. We individually evaluate each facility within simplified SPP and ERCOT footprints that have data available[7] for each of our five representative SMR types. To quantify the robustness of our results against uncertainties in technologies and economics, we run a subset of our facility processes (described in Table 2) with alternate historical wholesale power years, increased capital costs, constant thermal demand, reduced

flexibility, inclusion of TES, and the introduction of a carbon price (see Table 5 and supplemental information, section sensitivity analysis description).

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.joule.2023.03.009.

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AUTHOR CONTRIBUTIONS

M.V., methodology, software, analysis, data curation, writing – original draft, writing – review & editing, visualization; D.P., methodology; T.A., conceptualization, writing – review & editing, funding acquisition; D.C., conceptualization, methodology, funding acquisition; M.T.C., conceptualization, methodology, writing – review & editing, supervision, project administration, funding acquisition.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

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Joule, Volume 7

Supplemental information

Technoeconomic analysis of small modular reactors decarbonizing industrial process heat

Max Vanatta, Deep Patel, Todd Allen, Daniel Cooper, and Michael T. Craig

Supplemental Experimental Procedures & Items

Supplemental Experimental Procedures

SI.1 Model Formulation

SI Table 1: Definitions of model sets, variables, and parameters.

Set	**Definition**	**Indexed by**
G	SMR types (iPWR, iMWR, HTGR, PBR-HTGR, and microreactor)	g
N	Module within maximum module count	n
T	Hourly timestep	t
Variable	**Definition**	**Indexed across:**
πr	Annual revenue [$]Reactor output above minimum stable load (MWh)	-
c	Annualized costs [$]	-
s	Binary indicator of 'built' SMR type	G
m	Binary indicator of 'built' module	G,N
e	Electric generation [MW_e]	G,N,T
e'	Electric generation above the min. [MW_e]	G,N,T
r	Reactor core output [MW_{th}]	G,N,T
r'	Reactor core output above the min. [MW_{th}]	G,N,T
h	Process heat provided to industry [MW_{th}]	G,N,T
v	Commitment binary indicator	G,N,T
w	Decommitment binary variable	G,N,T
o	Commitment state, binary	G,N,T
oe	Electric generation commitment state, binary	G,N,T
Parameter	**Definition**	**Indexed across**
LMP	Hourly locational marginal price, day ahead market [$/$MWh_e$]	T

SC	Start-up cost for commitment [$/start]	G
VC	Variable costs including O&M and fuel costs [$/MWh$_{th}$]	G
FC	Fixed costs [$/MW$_e$/yr]	G
OCC	Overnight Capital Costs [$/MW$_e$]	G
CRF	Capital recovery factor	-
D	Scaling factor to account for length of optimization	-
IR	Interest rate [%/yr]	-
LT	Lifetime [yrs]	-
Cap	Nameplate capacity [MW$_e$]	G
Dem	Facility hourly demand [MW$_{th}$]	T
η_{TE}	Thermal efficiency [%]	G
η_{HT}	Heat transfer efficiency from core to working fluid [%]	G
TS	Thermal quality satisfaction binary	G
MSL	Minimum stable load [MW$_e$]	G
MSTL	Minimum stable turbine load [MW$_e$]	G
RR	Maximum ramp rate [MW$_e$/hr]	G
MDT	Minimum down time [hr]	G

To assess the technoeconomic value of SMRs in industrial heat applications, we design a profit-maximizing mixed integer linear program that optimizes on a per-industrial-facility-basis the number and type of SMR modules deployed and the hourly operation of those modules. Our model's objective function is to maximize annualized net revenues:

(1)

$$maximize\ \pi - c$$

Where revenues are defined as revenues from wholesale electricity sales:

(2

$$s.t. \qquad \pi = \sum_{g,n,t} e_{g,n,t} \times LMP_t \quad g \in G, n \in N, t \in T$$

And where costs equal annualized capital costs, plus SMR operational costs:

(3)

$$c = \sum_{g,n,t} \Big(\big(SC_g \times v_{g,n,t} \big) + \big(VC_g \times r_{g,n,t} \big) \Big) + \sum_{g,n} \big(Cap_g \times m_{g,n} \times (OCC_g \times CRF + FC_g) \big) \times D \quad g \in G, n \in N, t \in T$$

In which we annualize using a capital recovery factor (CRF) assuming a 30-year lifetime and a 7% interest rate [1].

(4)

$$CRF = \frac{IR}{1 - (1 + IR)^{-LT}}$$

Our model includes numerous constraints. First, we require heat output by the SMR to meet hourly industrial heat demand:

(5)

$$\sum_{g,n} \big(h_{g,n,t} \big) = Dem_t \quad g \in G, n \in N, t \in T$$

We require at least one module to be built at each facility. This module can be of any SMR type.

(6)

$$\sum_{g,n} m_{g,n} \geq 1 \quad g \in G, n \in N$$

Only one SMR type may be deployed at a given facility:

(7)

$$\sum_{g} s_g = 1 \quad g \in G$$

Our model can only build modules for the chosen SMR type:

(8)

$$m_{g,n} \leq s_g \quad g \in G, n \in N$$

Modules that are not built may not be committed, i.e. used for electricity or heat generation:

(9)

$$o_{g,n,t} \leq m_{g,n} \quad g \in G, n \in N, t \in T$$

We limit the electricity output of each module to its nameplate capacity:

(10)

$$e_{g,n,t} \leq Cap_g \quad g \in G, n \in N, t \in T$$

Additionally, each module reactor core output, converted to MW_e, must be less than the nameplate capacity multiplied by the commitment state:

(11)

$$r_{g,n,t} \times \eta_{HT_g} \times \eta_{TE_g} \leq Cap_g \times o_{g,n,t} \quad g \in G, n \in N, t \in T$$

Per module, the process heat supplied to industry plus the electricity generated, converted to MW_e, must not exceed the reactor core output with heat transfer losses:

(12)

$$h_{g,n,t} + \frac{e_{g,n,t}}{\eta_{TE_g}} \leq r_{g,n,t} \times \eta_{HT_g} \quad g \in G, n \in N, t \in T$$

Each module may only turn on or off, but not both in each timestep:
(13)

$$v_{g,n,t} + w_{g,n,t} \leq 1 \quad g \in G, n \in N, t \in T$$

Electricity generation may not be committed when the overall module is not committed:
(14)

$$oe_{g,n,t} \leq o_{g,n,t} \quad g \in G, n \in N, t \in T$$

And electric capacity is effectively zero when electric generation is not committed:
(15)

$$e_{g,n,t} \leq Cap_g \times oe_{g,n,t} \quad g \in G, n \in N, t \in T$$

We account for thermal quality requirements by only allowing SMR modules with sufficient working fluid temperatures to be committed:
(16)

$$o_{g,n,t} \leq TS_g \quad g \in G, n \in N, t \in T$$

The core generation above the minimum stable load (MST) is the reactor output minus the MSL, converted to MW_{th}, multiplied by the commitment state. Thus, the minimum when uncommitted is zero:
(17)

$$r'_{g,n,t} = r_{g,n,t} - \left(\frac{MSL_g}{\eta_{TE_g}} \times o_{g,n,t}\right) \quad g \in G, n \in N, t \in T$$

Similarly, the electric generation above minimum is the electric generation minus the minimum stable turbine load (MSTL) multiplied by the electricity commitment:
(18)

$$e'_{g,n,t} = e_{g,n,t} - \left(MSTL_g \times oe_{g,n,t}\right) \quad g \in G, n \in N, t \in T$$

We constrain ramping of reactor core output magnitude to be less than ramp rate parameter, converted to MW_{th}:
(19)

$$\left|r'_{g,n,t-1} - r'_{g,n,t}\right| \leq \frac{RR_g}{\eta_{TE_g}} \quad g \in G, n \in N, t > 0$$

Commitment of a module occurs when the commitment state between two consecutive time steps increases from 0 to 1 while the decommitment is 0:
(20)

$$v_{g,n,t} = o_{g,n,t} - o_{g,n,t-1} + w_{g,n,t} \quad g \in G, n \in N, t \in T$$

Decommitment of a module occurs when the commitment state between two consecutive time steps decreases from 1 to 0 while the commitment is 0:
(21)

$$w_{g,n,t} = o_{g,n,t-1} - o_{g,n,t} + v_{g,n,t} \quad g \in G, n \in N, t \in T$$

Commitment may only occur if the sum of previous time steps during the length of the minimum down time (MDT) is greater than the MDT:
(22)

$$\sum_{i=1}^{MDT_g} \left(1 - o_{g,n,t-i}\right) \geq v_{g,n,t} \times MDT_g \quad g \in G, n \in N, t \in T$$

SI.2 Thermal Energy Storage (TES) Constraints and Data

We test the sensitivity of our results to deployment of TES alongside SMRs. Given its technological maturity, we specifically consider a molten nitrate salt-based TES system using values from [2]. While molten nitrate salts might not be capable of serving demands above >565 °C, it is solely a stand-in for a more in-depth analysis of TES technologies in future work. For each industrial facility, we model a single TES system that all SMR modules can charge. When discharging, the TES system only serves process heat demand. While the power and efficiency of TES will vary with temperature and temperature difference, we assume a linear behavior with a simple energy exchange according to the following balance equations and constraints.

We constrain the state of charge (SOC) to be less than the TES capacity:

(23)

$$l_{g,t} \leq Cap_g^{TES} \quad g \in G, t \in T$$

And the SOC is equal to the SOC at the previous time step minus latent losses, minus discharged heat, and plus charged heat from each module:

(24)

$$l_{g,t} = l_{g,t=1} \times \eta_L - h_{g,t-1}^D + \sum_n h_{g,n,t-1}^C \quad g \in G, n \in N, t \in T$$

The discharge rate is limited by the TES maximum charge/discharge rate:

(25)

$$h_{g,t}^D \leq H_g \quad g \in G, t \in T$$

And the charging across all modules may not exceed this maximum charge/discharge rate:

(26)

$$\sum_n h_{g,n,t}^C \leq H_g \quad g \in G, n \in N, t \in T$$

Where superscript 'TES' indicates TES system parameters; l = TES system state of charge (SOC) [MWh_{th}];Cap^{TES} = TES capacity [MWh_{th}]; η_L = self-discharge rate [%]; h^D= TES discharge [MWh_{th}]; h^C = TES charge [MWh_{th}]; H = TES maximum charge/discharge rate [MWh_{th}/hr]. It should also be noted, we apply the charge and discharge efficiencies in the reactor energy balance and industrial process heat balance for charge and discharge respectively.

Additionally, constraints shown in *Eqs. 3, 5,* and *12* need to be updated to included TES such that the capital costs now include the TES capital costs:

(27)

$$c_t = \sum_{g,n} \left((SC_g \times v_{g,n,t}) + (VC_g \times r_{g,n,t}) \right) + \sum_{g,n} \left(Cap_g \times m_{g,n} \times (CC_g \times CRF + FC_g) + \boldsymbol{Cap_g^{TES}} \times (\boldsymbol{CC_g^{TES}} \times \boldsymbol{CRF}) \right) \times D \quad g \in G, n \in N, t \in T$$

And the process heat demand is now satisfied by the sum of direct heat from the SMRs and TES discharge with discharge losses:

(28)

$$\sum_{g}\left(\sum_{n}\left(h_{g,n,t}\right)+\boldsymbol{h^{D}_{g,t}}\times\boldsymbol{\eta_{D}}\right)=\ Dem_{t}\quad g\in G,n\in N,t\in T$$

The sum of process heat, electric generation and TES charging may not exceed the reactor core output:

(29)

$$h_{g,n,t}+\frac{e_{g,n,t}}{\eta_{TE}}+\frac{\boldsymbol{h^{C}_{g,n,t}}}{\boldsymbol{\eta_{C}}}\leq r_{g,n,t}\times\eta_{HT}\quad g\in G,n\in N,t\in T$$

Our TES parameters are shown in *Table 1*. We assume a high self-discharge rate of 1% loss per hour, though this value is highly dependent on system design, though McLauchlan et al. [3] suggest that with as low as 0.2m insulation thickness the self-discharge rate would be 1% per day for a 300° C temperature difference. We test three TES system sizes of 50%, 100%, and 200% of peak thermal demand.

SI Table 2: TES parameters

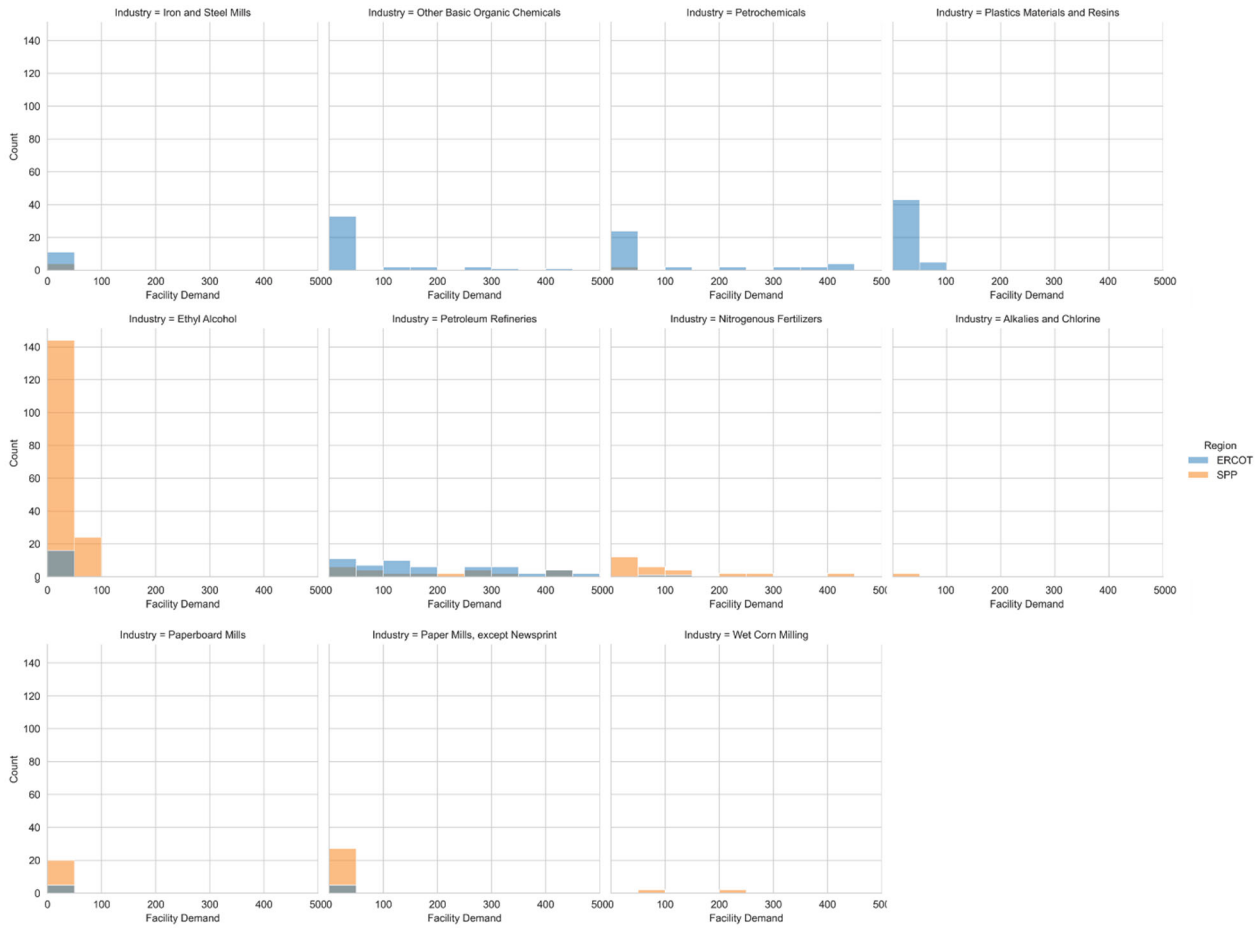
	Discharge Efficiency: η_{DIS}	**Charge Efficiency: η_{CHG}**	**Self-discharge Rate: η_{L}**	**Capital Cost [$/kWh$_t$]**
Molten Nitrate Salt	99% [2]	99% [2]	1%	20 [2]

SI.3 Sensitivity Analysis Description

We perform sensitivity analyses for multiple uncertainties. For discussion of results, see main text result section *Sensitivity Analyses* and section *SI.9 Sensitivity Results*. In each case, the overall model formulation remains the same and only the specified input parameters for each sensitivity analysis are changed (Table 5). In the market year sensitivity for example, only the power prices available during joint heat and wholesale market participation will be replaced. Similarly, for the capital cost uncertainty and reactor core flexibility, only the input parameters are adjusted as described in SI Table 3. For the industrial thermal demand variability, we operated the model as before, but when the NAICS code for an industry would have otherwise been associated with an hourly variation, it was instead assumed to be constant. Carbon pricing was solely applied post optimization by multiplying the avoided natural gas by its emission factor and the given carbon pricing level.

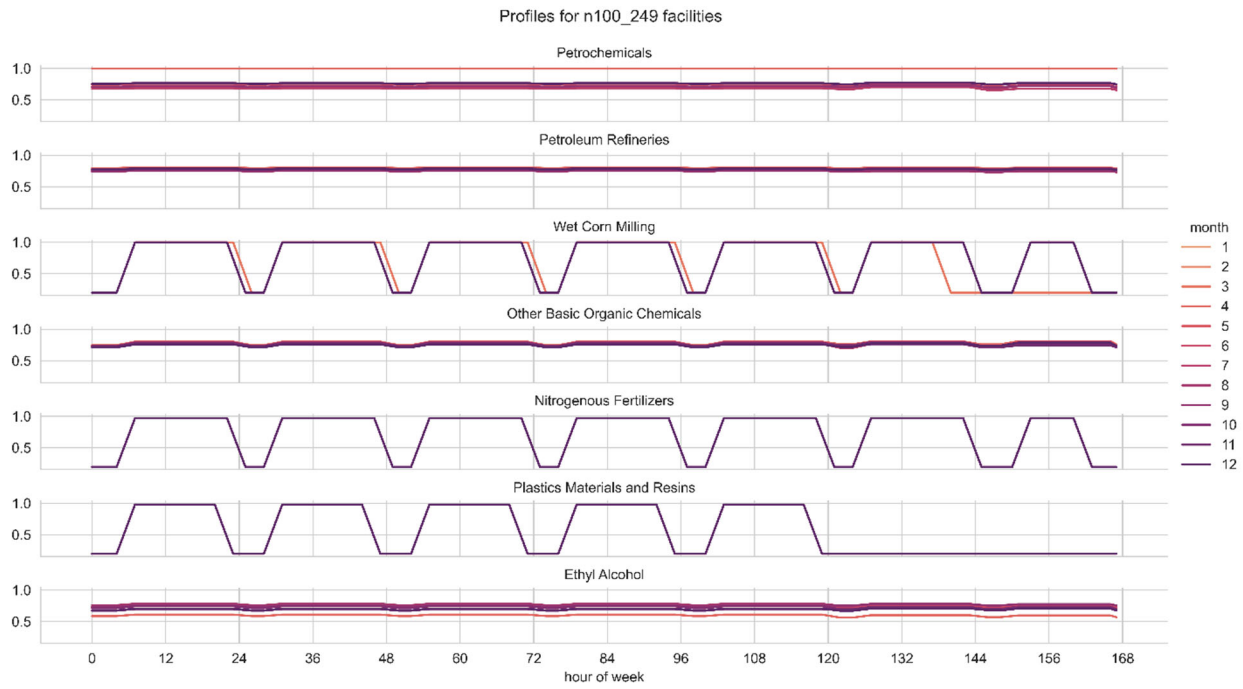
SI.4 Industrial Facility Characteristics

Using data from [4], we isolate the temperature ranges and fuel types which are present in the 12 industries across ERCOT and SPP (Figure 1). To increase visual clarity, we have grouped fuels by their general types. Our analysis only considers process heat demand served by natural gas, which Figure 1 illustrates are the dominant fuel type. Additionally, we have plotted the distribution of individual facility demands for each region and industry type (SI Figure S1). The majority of these industries are dominated by small scale, < 50 MW_{th}, demands. Petroleum refineries and nitrogenous fertilizer demands are the most widely distributed industry types.



SI Figure S1: Distribution of facilities by their average demand. SPP is in blue and ERCOT is in orange. The facilities are skewed towards low demands.

For some industries, hourly thermal demand estimations are available from McMillan et al. (2019) [5]. This hourly data is available for one week per month. All other industries are assumed to have constant demand.

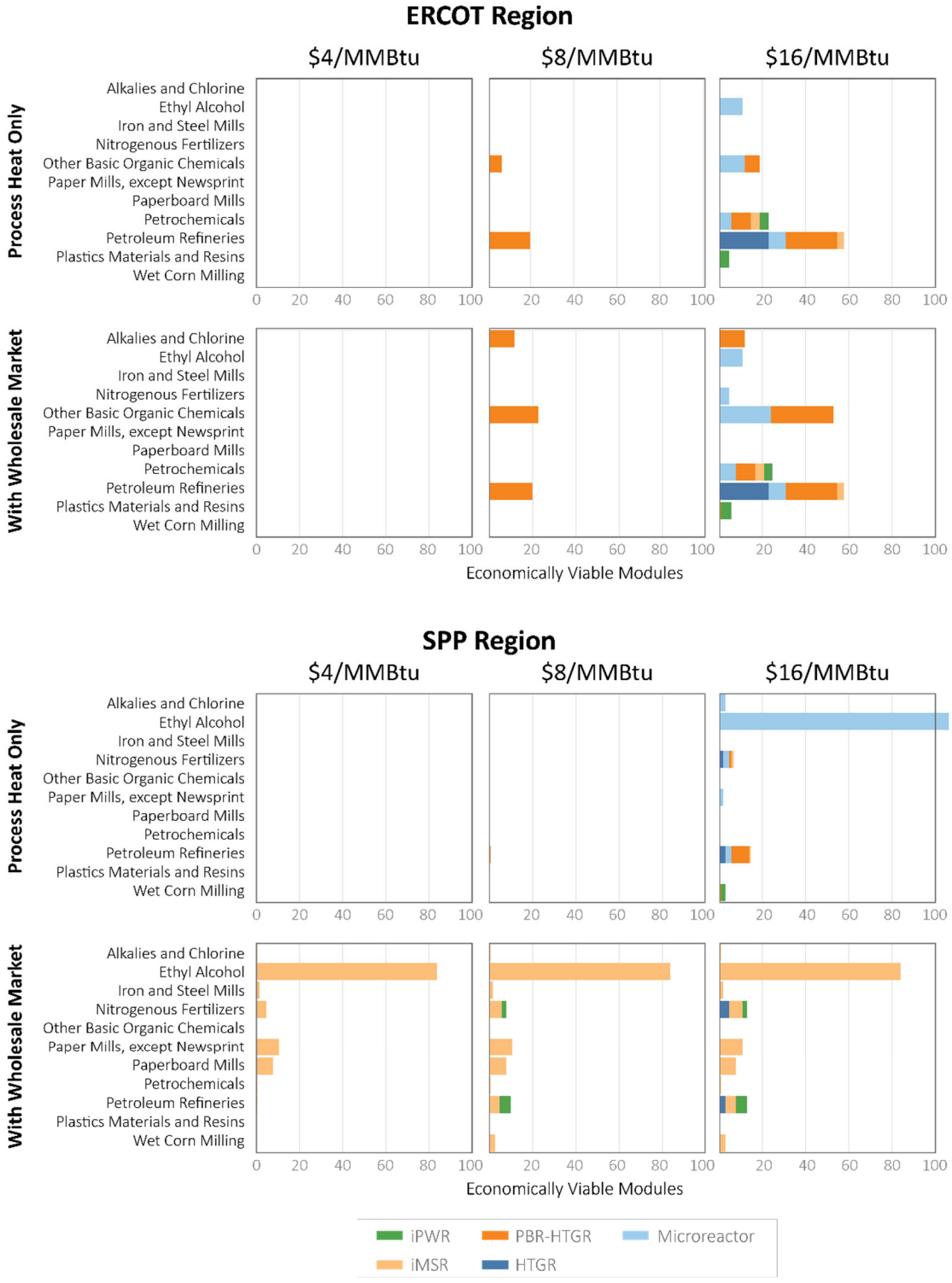


SI Figure S2: Normalized hourly thermal demands by industry and month, when available. Data gathered from [5]*. Hourly profiles vary by the workforce size. This selected range, 100 to 249 employees, demonstrates both the hourly variations of the largest facilities, such as petrochemical and petroleum refineries, as well as smaller facilities in our study set, such as wet corn milling.*

Supplemental Items

SI.5 Per-SMR Type Deployment By Industry

As an extension to main text Figure 4, we find the economic viability of SMRs in each industry at three representative natural gas prices (SI Figure S3). This stacked bar representation clearly shows the early deployment of PBR-HTGRs in ERCOT, followed by a widely mixed selection by $16/mmbtu. In SPP, providing process heat only, microreactors dominate deployment, while iMSRs dominate when facilities participate in wholesale power markets.



SI Figure S3: SMR economic feasibility per industry type. Left column is $4/MMBtu of avoided natural gas, middle is $8/MMBtu, and right is $16/MMBtu. Rows from the top down are ERCOT process heat only, ERCOT joint participation in power and heat markets, SPP process heat only, and SPP joint participation in power and heat markets.

SI.6 Operational Characteristics of SMRs

In each scenario, we find the operational characteristics such as capacity factor, electricity production, etc. to better understand how SMRs are utilized in these contexts. We report the capacity factor (percent of total possible usage of the SMR across the entire year), electricity usage (percent of total energy produced by the SMR for generating electricity for wholesale power markets), reactor usage (total energy output from the SMR), and the electricity produced across the system, peak, and then per facility statistics (SI Table S3).

SI Table S3: Regional and per-facility operational characteristics of SMRs in various market conditions. Median is labelled 'Med.' and standard deviation is 'std. dev.'. If a field has a dash, it indicates that this is not a tested condition or that the field is not measurable (such as std. dev. for SPP process heat only at $8/MMBtu in which there is only one facility).

	Breakeven [$/MMBtu]		Capacity Factor [%]			Electricity Usage [%]			Reactor Usage				Electricity Produced				
										Per-Facility [MW-th]					Per-Facility [MW-e]		
			Med.	Mean	Std. Dev.	Med	Mean	Std. Dev.	Total [TWh_{th}]	Med.	Mean	Std. Dev.	Total [TWh_e]	System Peak [GW_e]	Med.	Mean	Std. Dev.
ERCOT	4	Process Heat Only	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		w/ Wholesale	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	8	Process Heat Only	53	49	9	-	-	-	40.54	186.05	171.41	32.86	-	-	-	-	-
		w/ Wholesale	56	56	1	9	10	8	93.82	194.73	194.73	2.26	4.32	.84	8.31	8.96	7.01
	16	Process Heat Only	67	70	28	-	-	-	140.76	137.48	138.52	108.16	-	-	-	-	-
		w/ Wholesale	97	80	22	18	21	4	242.22	192.85	162.65	111.36	21.37	3.54	8.45	14.35	15.92
SPP	4	Process Heat Only	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		w/ Wholesale	97	97	0	89	87	7	287.48	290.39	290.41	0.09	119.65	14.83	123.22	120.87	9.02
	8	Process Heat Only	51	51	-	-	-	-	1.57	179.11	179.11	-	-	-	-	-	-
		w/ Wholesale	97	97	0	89	84	15	316.42	290.39	288.97	8.31	125.77	15.78	122.56	114.86	22.68
	16	Process Heat Only	100	89	21	-	-	-	50.92	20.00	41.52	62.95	-	-	-	-	-
		w/ Wholesale	97	97	0	88	81	18	340.15	290.39	291.96	14.35	130.38	16.62	121.90	111.91	26.25

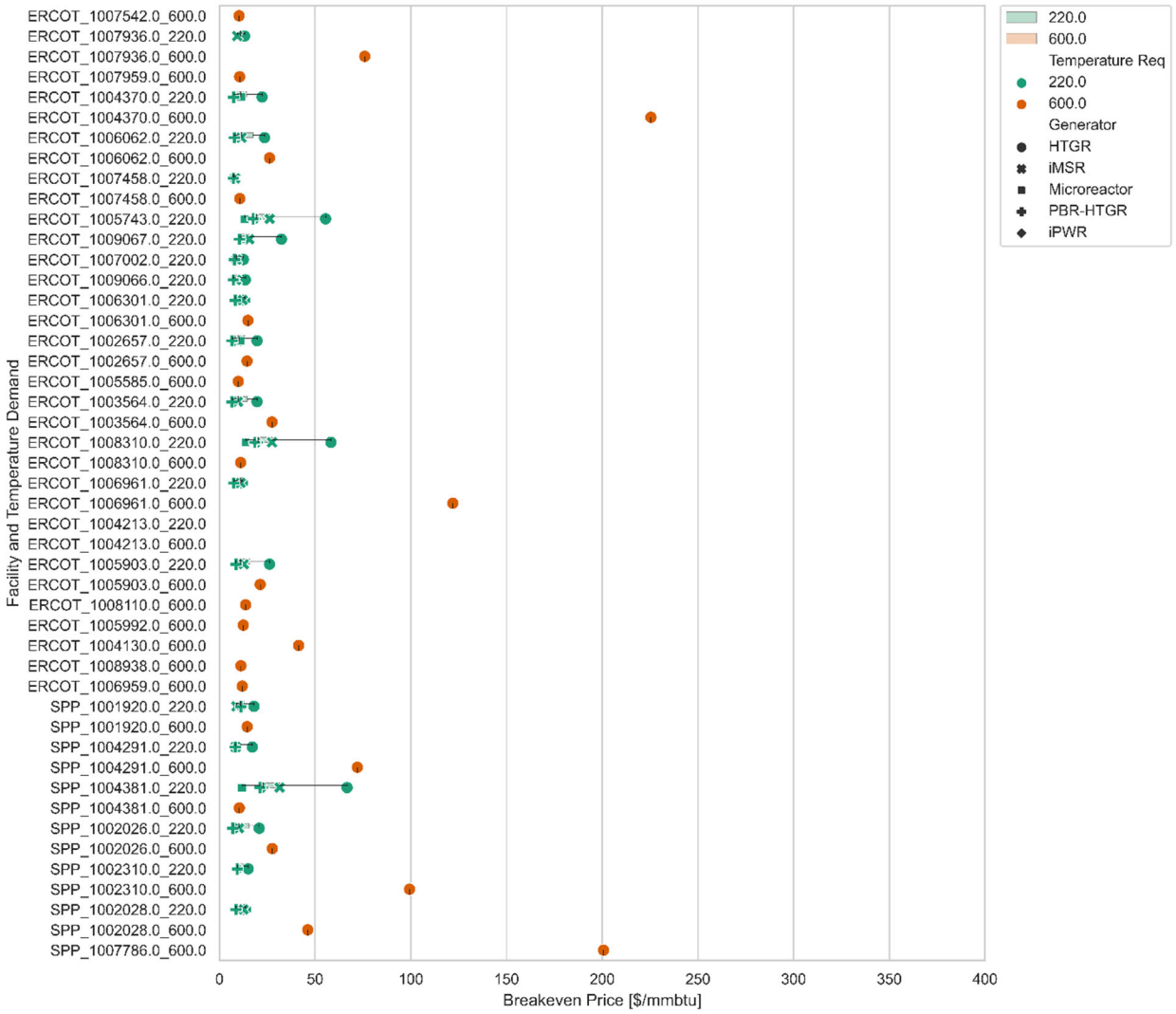
SI.7 Per-Facility, Per-SMR Type Breakeven Prices

We modeled the operations and economics of each SMR type at each facility to better understand the dominance of deployment of particular SMR types in the least-cost model. For each facility, we plot the breakeven price for SMRs providing process heat only (SI Figures S4-S14). Some industries such as petroleum refineries have capacities and thermal quality requirements which allow them to capture the lowest LCOH regions of multiple SMR types (see main text Figure 3), resulting in a narrow range of breakeven natural gas prices for multiple SMR types. Other industries, such as plastics and resin manufacturing and iron and steel mills have a wide range between best and worst economically performing SMR types.

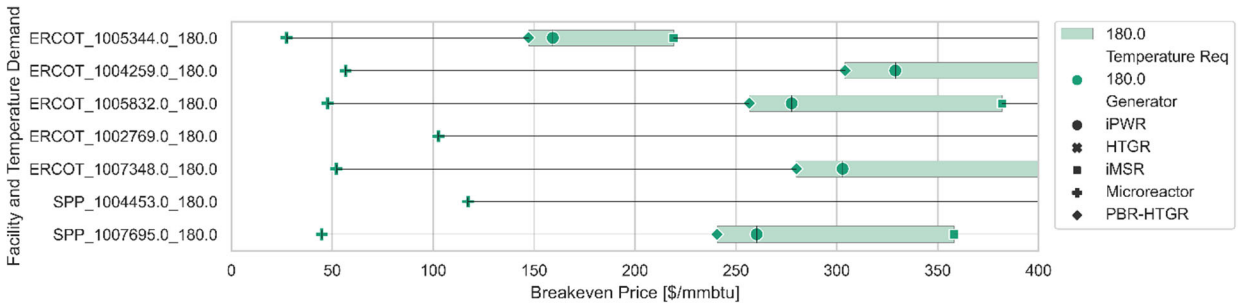
The range of the results on a per-facility basis is valuable in qualitatively incorporating the impact of SMR design and industrial facility integration uncertainty into our analysis that we do not quantitatively capture . In the

cases where the range between the most and less economically viable SMRs is substantial, e.g. greater than $200/MMBtu as seen in 291 °C processes (SI Figure S7), the likelihood that uncertainty will lead to a different SMR type becoming the most economically viable is low. In the cases where breakeven prices across SMR types is low, such as in petroleum refineries (SI Figure S4), the most economically viable SMR type and our modeled SMR investment decision is more sensitive to uncertainties

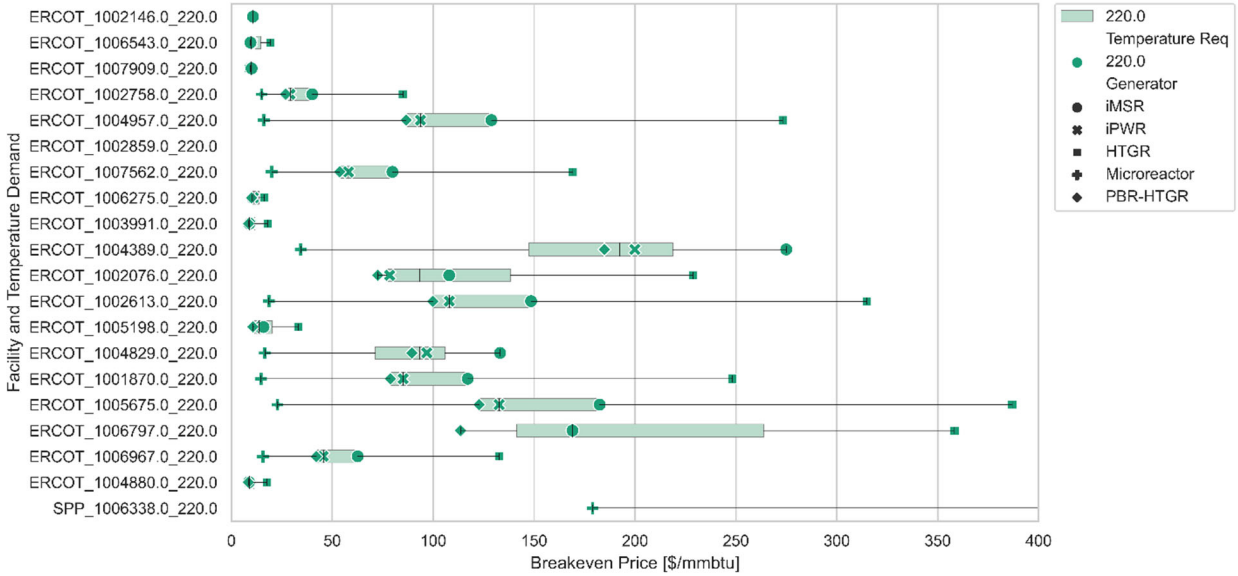
In addition to the capacity matching in defining the economic viability, the number of technologically feasible SMR types are limited based on the outlet temperatures and industrial heat temperature requirements. For example, only the HTGR breakeven price is available for processes at petroleum refineries above 600° C (hence only a single point is plotted for these processes in SI Figure S4).



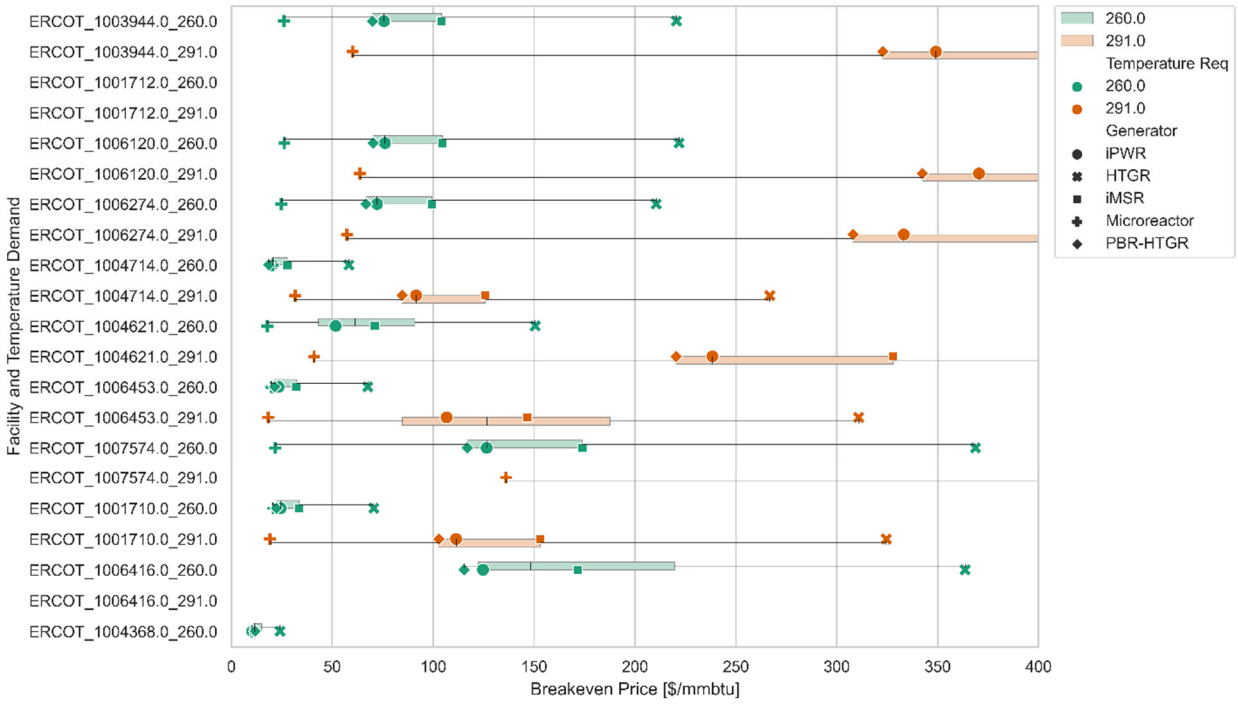
SI Figure S4: Petroleum refineries: Per-facility distribution of breakeven natural gas prices for each SMR type. Each facility/process is labelled as region_facility ID number_temperature required. Each color represents a temperature demand, and each point style indicates an SMR type (circle= HTGR, square= iMSR, X= microreactor, + = PBR-HTGR, and diamond = iPWR). The distribution of breakeven natural gas prices within this industry are tightly distributed compared to other industries.



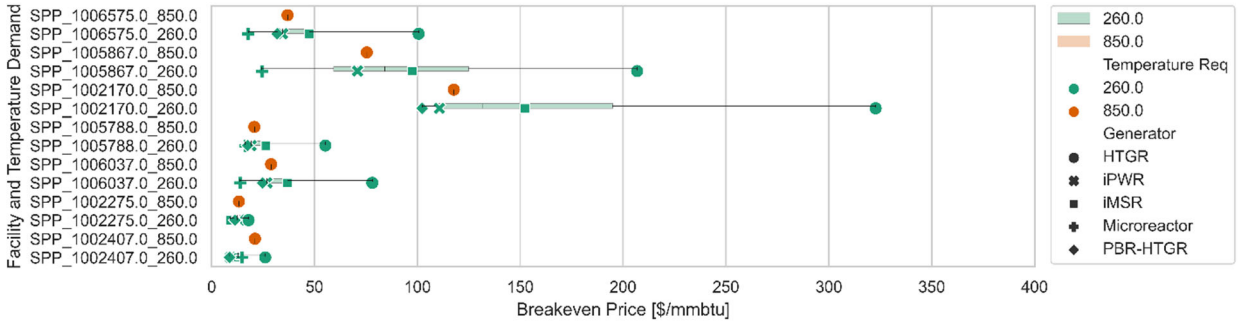
SI Figure S5:Iron and steel mills: Per-facility distribution of breakeven natural gas prices for each SMR type.



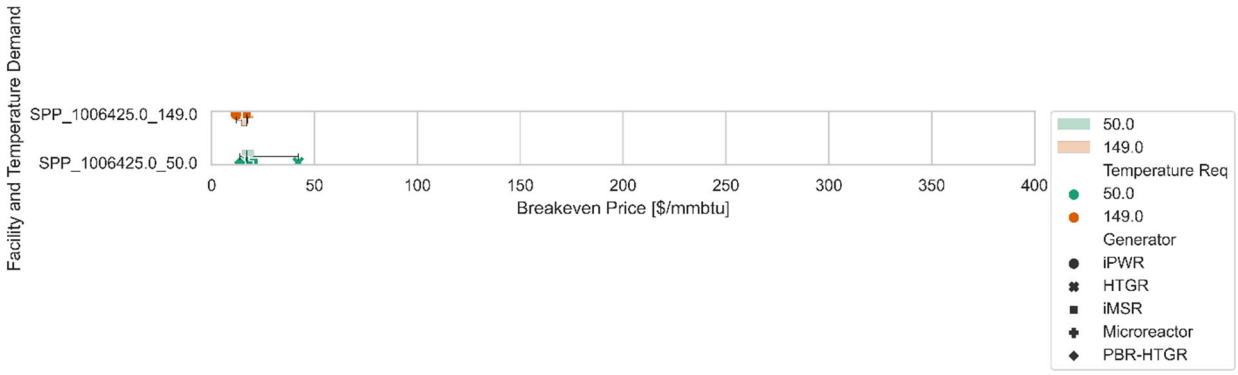
SI Figure S6:Other basic organic chemicals: Per-facility distribution of breakeven natural gas prices for each SMR type.



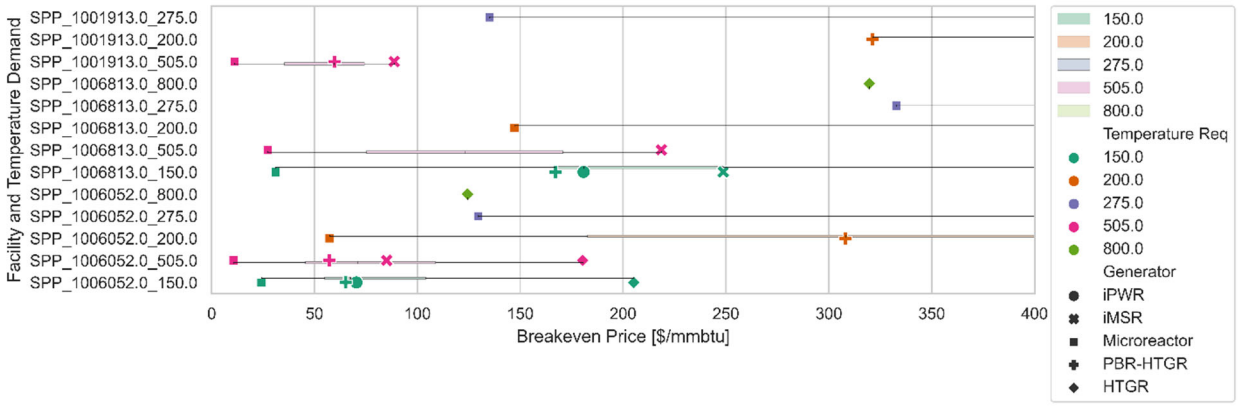
SI Figure S7: Plastic materials and Resins: Per-facility distribution of breakeven natural gas prices for each SMR type.



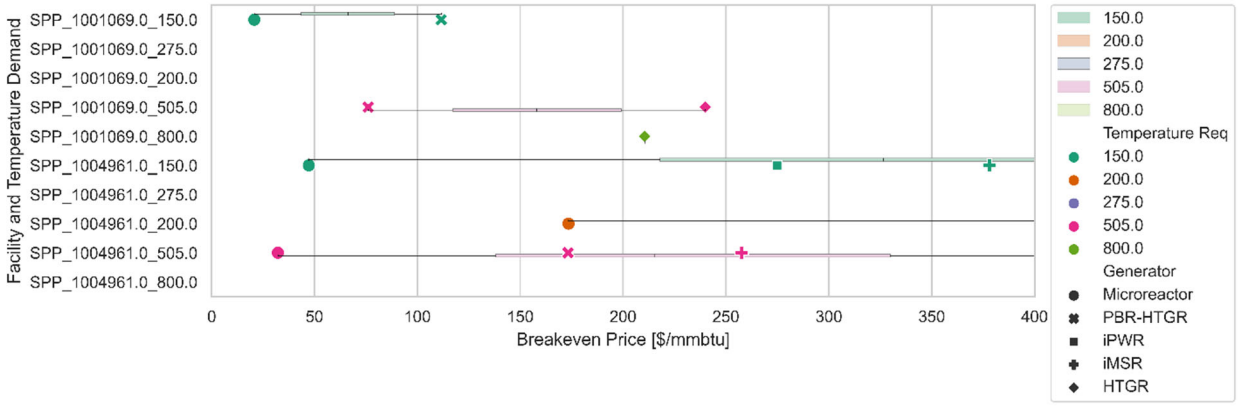
SI Figure S8: Nitrogenous fertilizers: Per-facility distribution of breakeven natural gas prices for each SMR type.



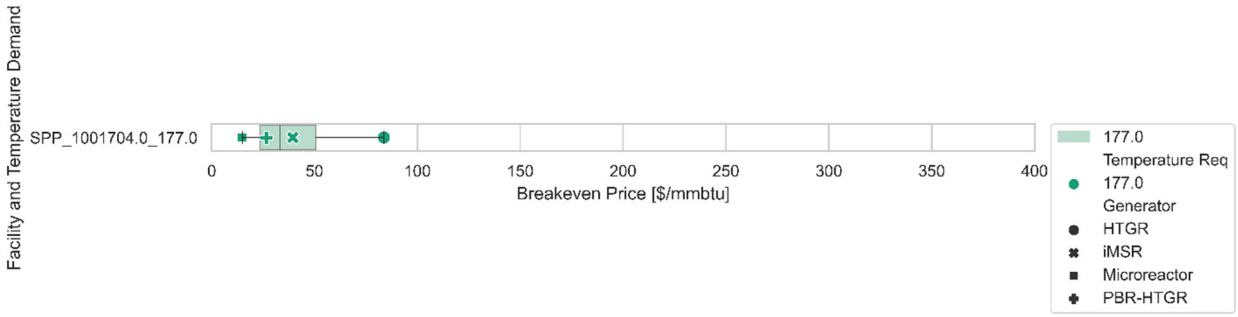
SI Figure S9:Wet corn milling: Per-facility distribution of breakeven natural gas prices for each SMR type.



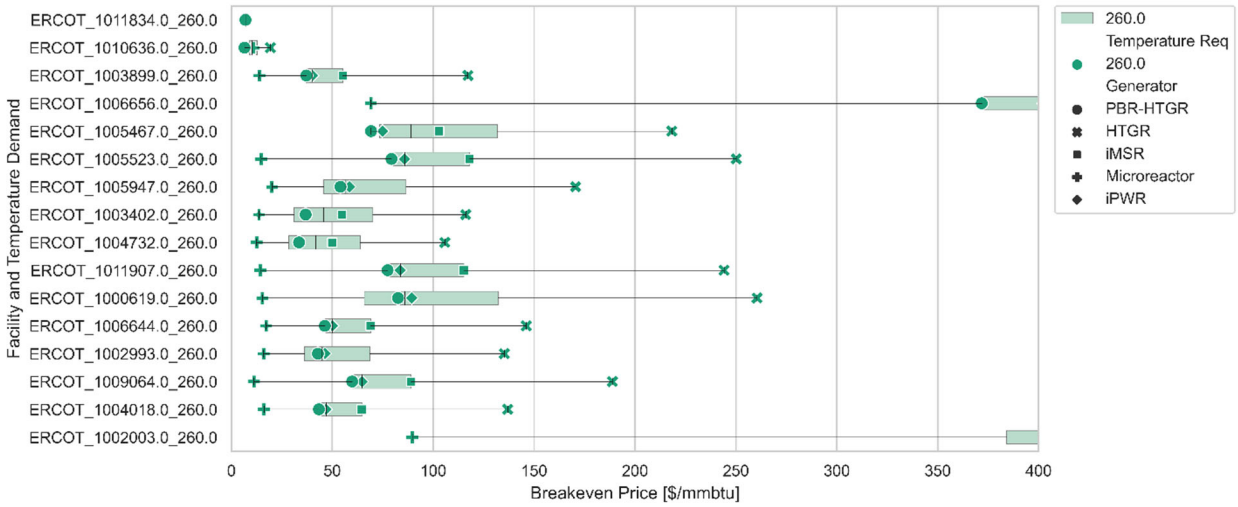
SI Figure S10:Paper mills, except newsprint: Per-facility distribution of breakeven natural gas prices for each SMR type.



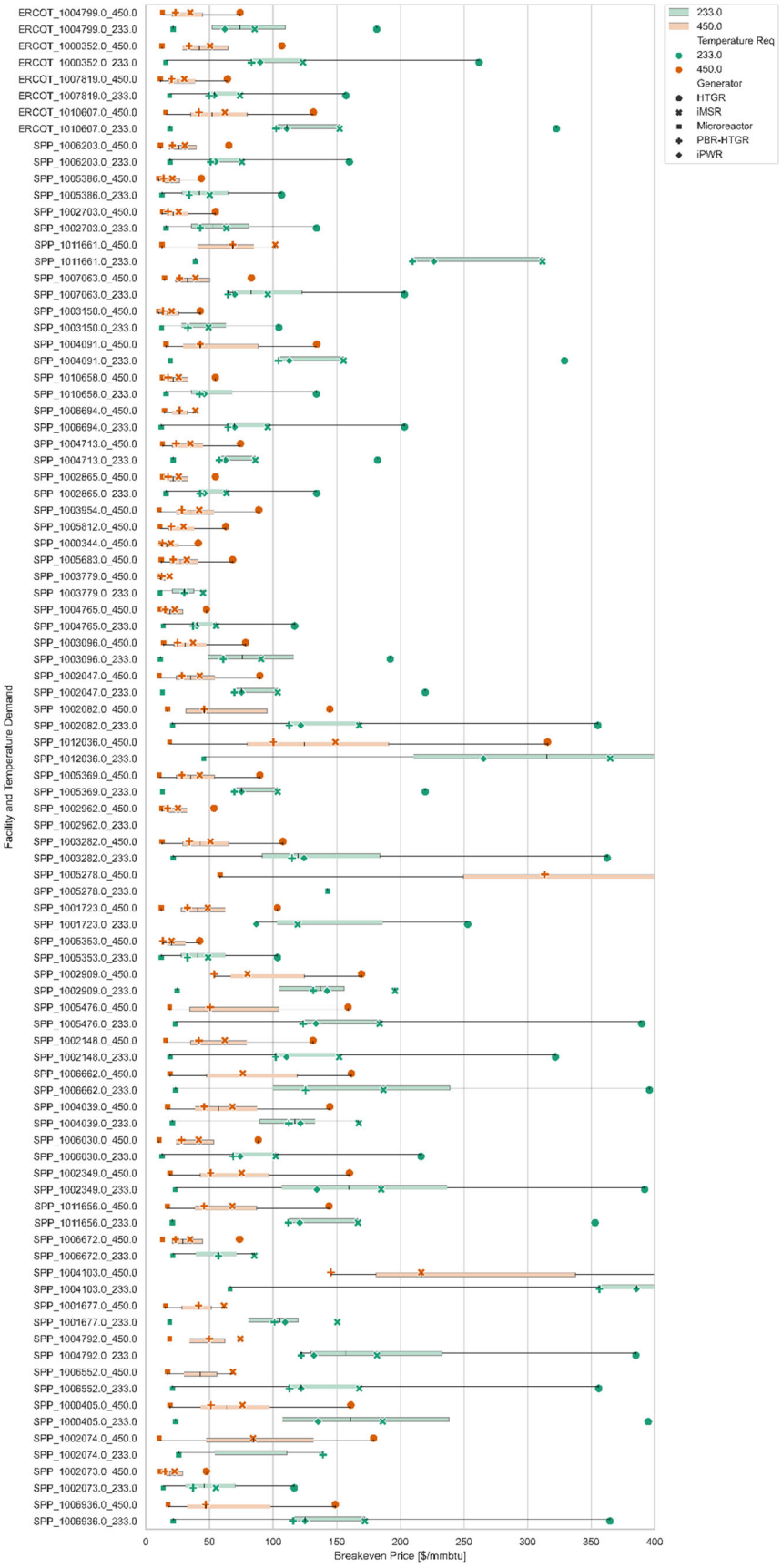
SI Figure S11:Paperboard mills: Per-facility distribution of breakeven natural gas prices for each SMR type.



SI Figure S12:Alkalies and Chlorine: Per-facility distribution of breakeven natural gas prices for each SMR type.



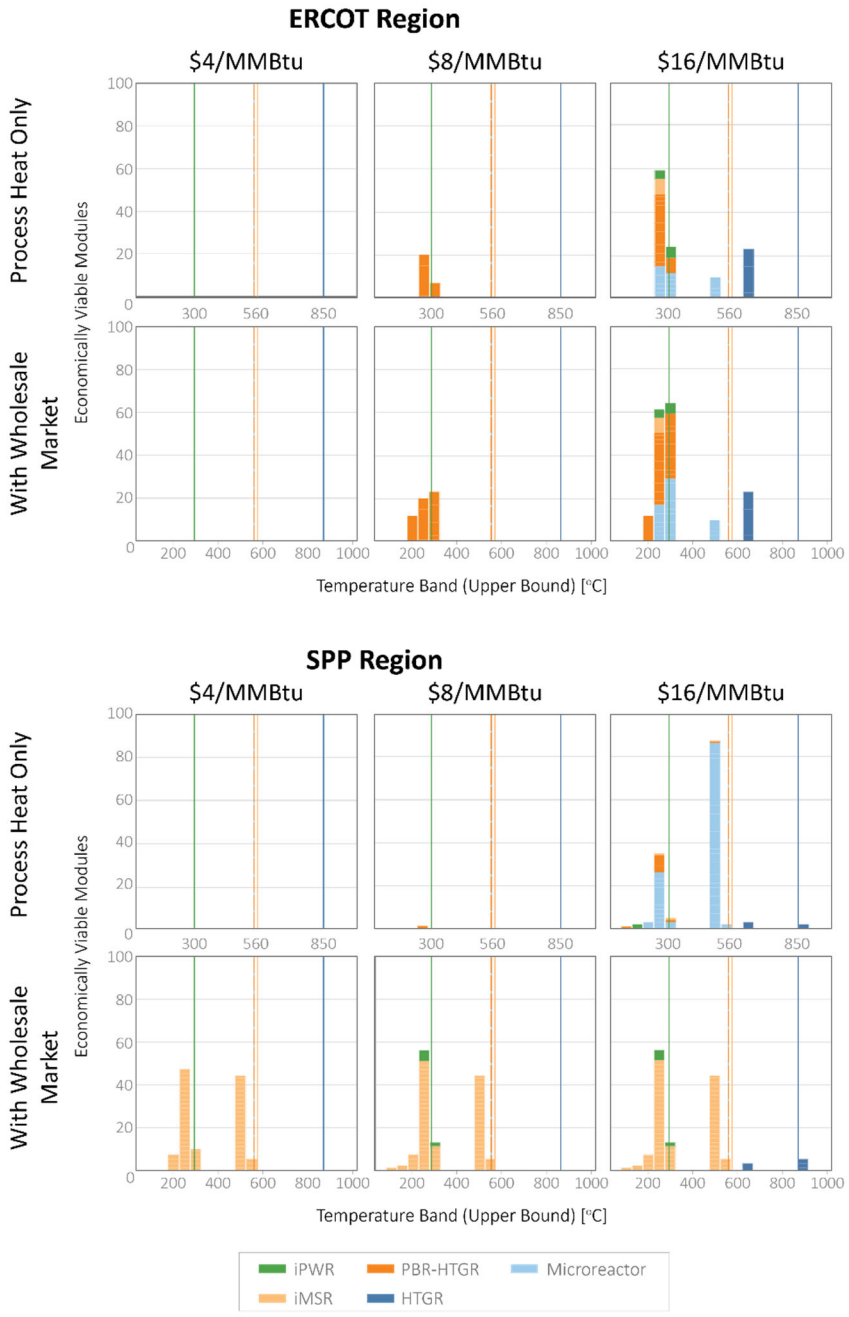
SI Figure S13:Petrochemicals: Per-facility distribution of breakeven natural gas prices for each SMR type.


ERCOT_1004799.0_450.0
ERCOT_1004799.0_233.0
ERCOT_1000352.0_450.0
ERCOT_1000352.0_233.0
ERCOT_1007819.0_450.0
ERCOT_1007819.0_233.0
ERCOT_1010607.0_450.0
ERCOT_1010607.0_233.0
SPP_1006203.0_450.0
SPP_1006203.0_233.0
SPP_1005386.0_450.0
SPP_1005386.0_233.0
SPP_1002703.0_450.0
SPP_1002703.0_233.0
SPP_1011661.0_450.0
SPP_1011661.0_233.0
SPP_1007063.0_450.0
SPP_1007063.0_233.0
SPP_1003150.0_450.0
SPP_1003150.0_233.0
SPP_1004091.0_450.0
SPP_1004091.0_233.0
SPP_1010658.0_450.0
SPP_1010658.0_233.0
SPP_1006694.0_450.0
SPP_1006694.0_233.0
SPP_1004713.0_450.0
SPP_1004713.0_233.0
SPP_1002865.0_450.0
SPP_1002865.0_233.0
SPP_1003954.0_450.0
SPP_1005812.0_450.0
SPP_1000344.0_450.0
SPP_1005683.0_450.0
SPP_1003779.0_450.0
SPP_1003779.0_233.0
SPP_1004765.0_450.0
SPP_1004765.0_233.0
SPP_1003096.0_450.0
SPP_1003096.0_233.0
SPP_1002047.0_450.0
SPP_1002047.0_233.0
SPP_1002082.0_450.0
SPP_1002082.0_233.0
SPP_1012036.0_450.0
SPP_1012036.0_233.0
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SPP_1001723.0_233.0
SPP_1005353.0_450.0
SPP_1005353.0_233.0
SPP_1002909.0_450.0
SPP_1002909.0_233.0
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SPP_1002148.0_233.0
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SPP_1006662.0_233.0
SPP_1004039.0_450.0
SPP_1004039.0_233.0
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SPP_1002349.0_233.0
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SPP_1006552.0_233.0
SPP_1000405.0_450.0
SPP_1000405.0_233.0
SPP_1002074.0_450.0
SPP_1002074.0_233.0
SPP_1002073.0_450.0
SPP_1002073.0_233.0
SPP_1006936.0_450.0
SPP_1006936.0_233.0
Facility and Temperature Demand
0
50
100
150
200
250
300
350
400
Breakeven Price [$/mmbtu]
233.0
450.0
Temperature Req
233.0
450.0
Generator
HTGR
iMSR
Microreactor
PBR-HTGR
iPWR

SI Figure S14:Ethyl alcohol: Per-facility distribution of breakeven natural gas prices for each SMR type.

SI.8 Economically Viable Deployments per-Temperature

We disaggregate the economic viability of SMRs by temperature demanded and the natural gas price (SI Figure S15). SMRs are only economically viable in SPP with wholesale power market participation.

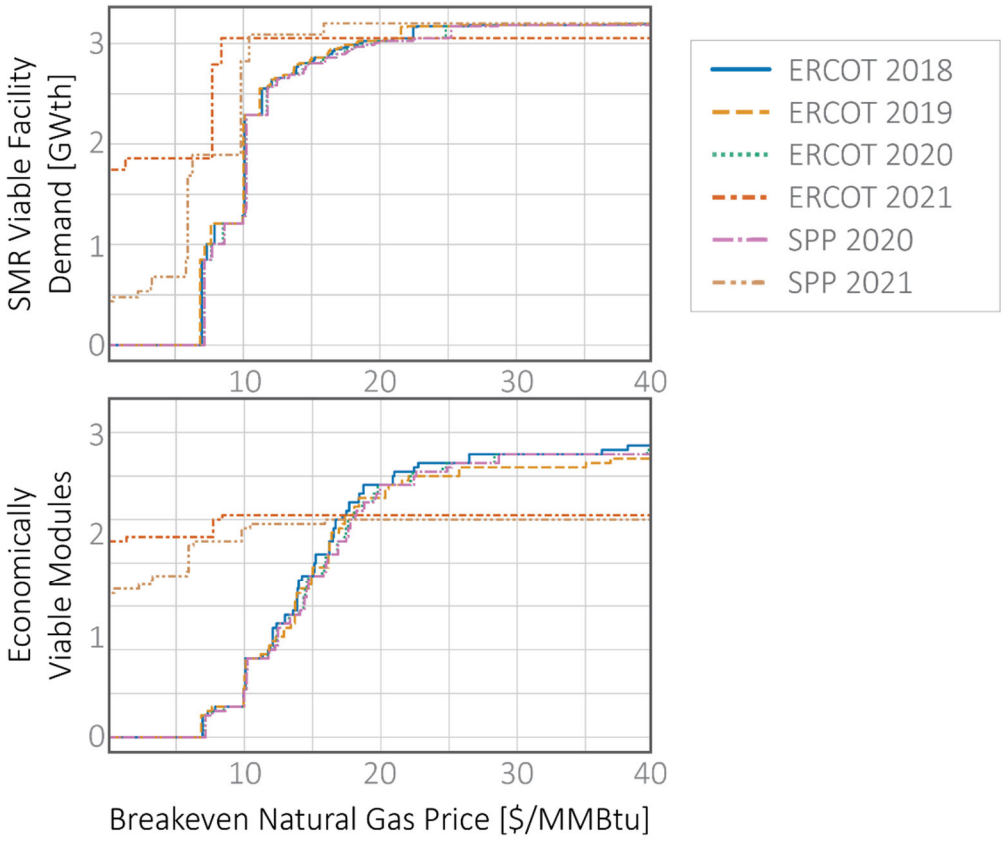


SI Figure S15: Number of economically feasible modules of each SMR type across temperature demanded at each representative natural gas price. In each sub graph, the upper bound temperature of each SMR type is marked with a line of the same color as the SMR bars. In all cases, the highest temperatures require the highest natural gas prices to become economically feasible with an SMR.

SI.9 Sensitivity Results

We tested the robustness of our results against multiple uncertainties in our parameters on a reduced set of 50 facility demands (subset descriptive statistics in Table 2). Flat industrial demands, varying capital costs, addition of TES to the SMR facility, and inflexible SMR operations have no substantial impact on deployment (Figure 6). Neglecting the variable thermal demands by assuming a constant thermal demand across the entire year does not alter the economic viability in the form of breakeven price of the number of modules. An increase by 10% in capital costs reduces the number of economically viable modules most in the process heat only scenario, by up to 12% across natural gas prices. Forcing SMRs to operate at 80% of capacity or greater does not change our results. Including TES in the system does not improve the SMR's potential in any scenario, though this system requires further investigation.

All lower LMP price years (2018-2020) result in similar SMR deployments (SI Figure S16). In 2021 in SPP, a high price year, we find that some facilities reach positive net revenues through wholesale market participation as described in the main text. In 2021 in ERCOT, an even higher price year, the effects are more exaggerated.



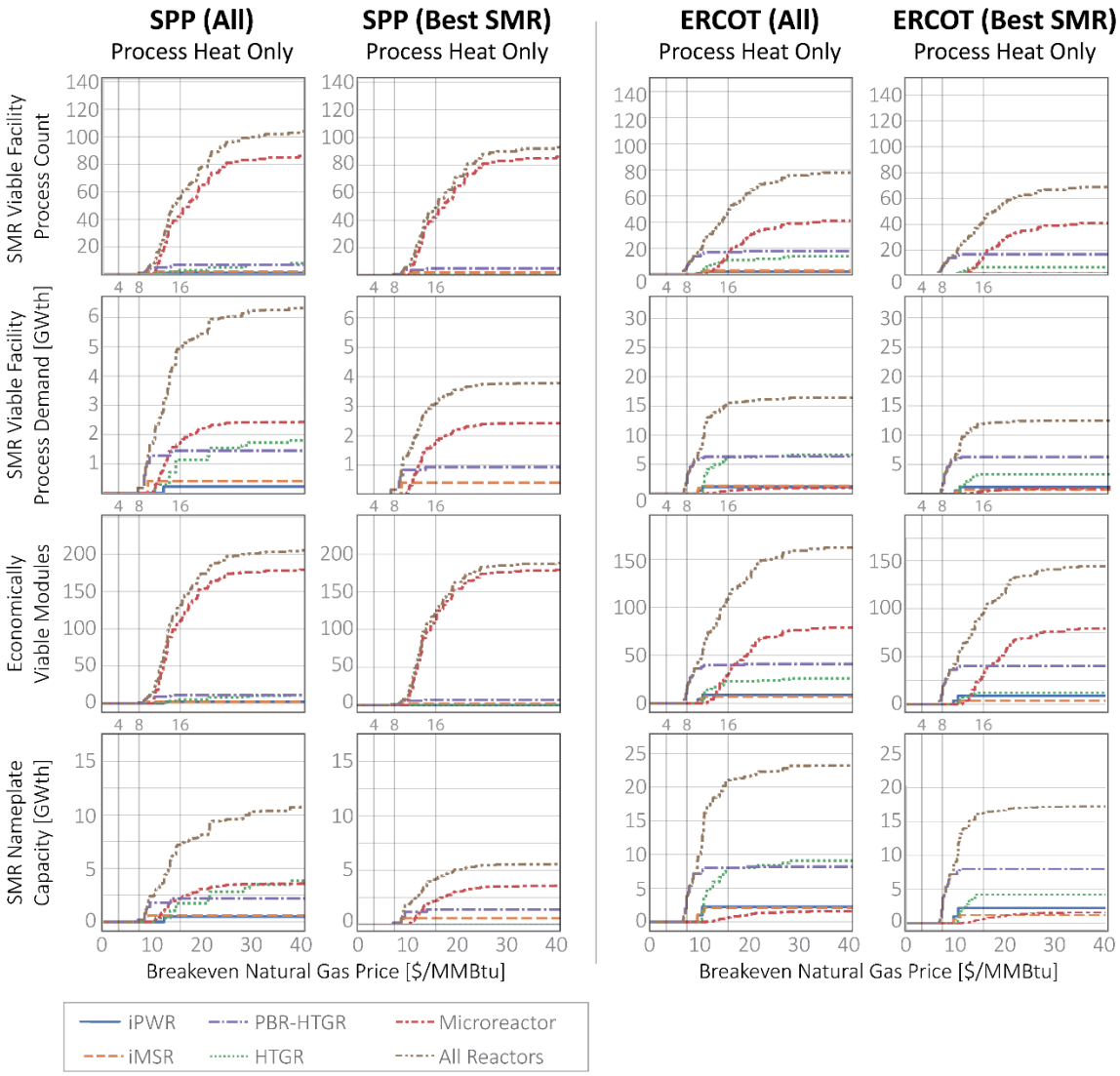
SI Figure S16: Facility demand (top) and module count (bottom) supply curves across multiple DAM years.

SI.10 SMR Type Limited Results

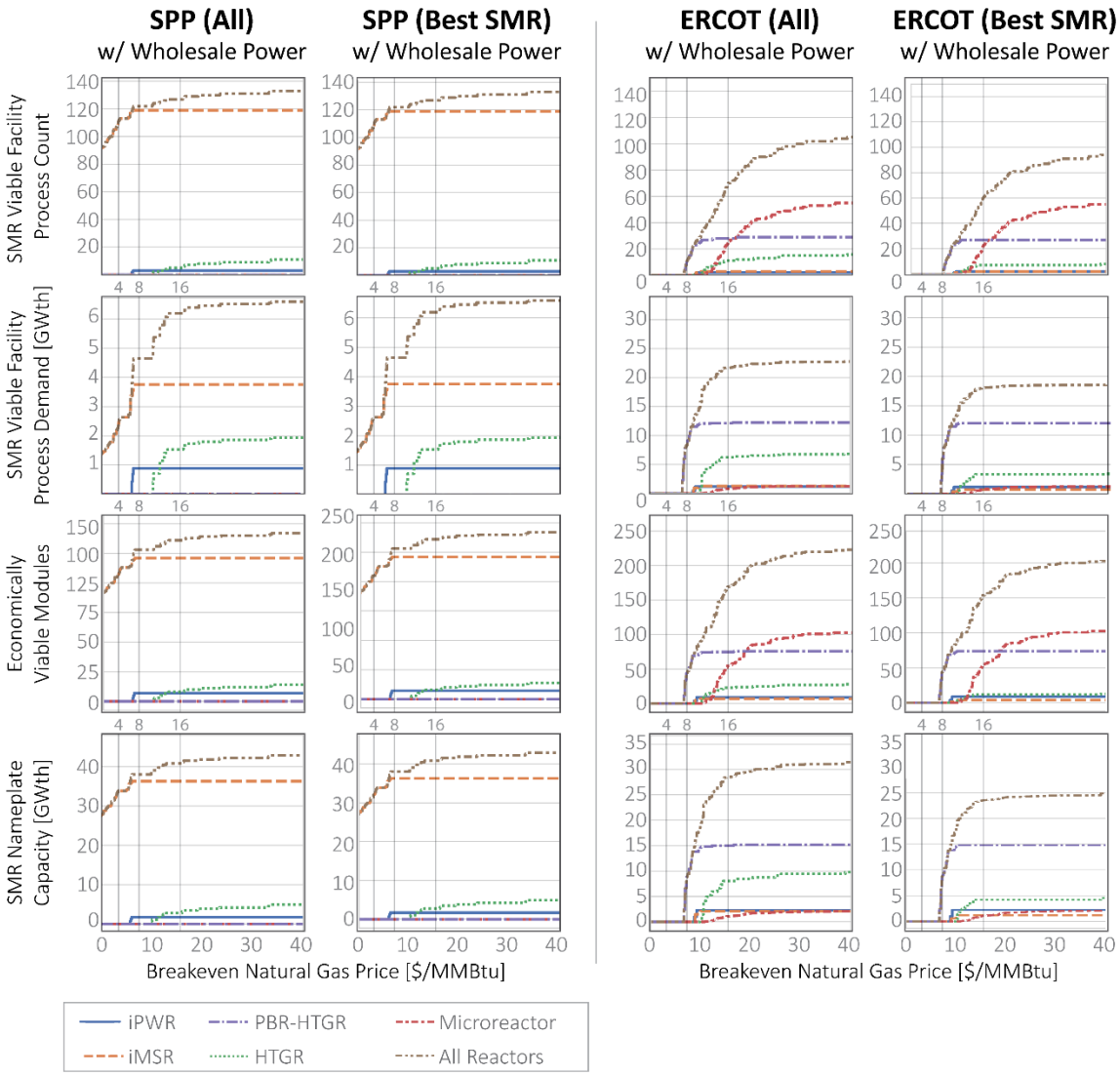
We perform our analysis on a per-facility-process level which allows the siting of two different SMR types within a single facility. This decision on the part of the facility may be challenging given the likelihood of partnerships in the early stages of SMR deployment, limiting the investment decision to single SMR type per facility limitations. Therefore, we compare the per-facility-process results presented in the main text to the deployment opportunities with a single SMR type per facility limit, i.e. only the most economical SMR type across all facility processes is allowed at a given facility. Multiple facility processes of different temperatures using the same SMR type are allowed.

We find that when serving only process heat, the number of facility processes and the number of deployed modules is only reduced between 5-15% across regions and breakeven costs (SI Figure S17). The economically viable facility process demand and SMR capacity decreases by roughly 20% and 50% in ERCOT and SPP respectively. This, along with the SMR-type specific curves shows that the larger, more expensive, higher temperature HTGR is not as widely deployable when the limit is in place. PBR-HTGRs, iMSRs, and iPWRs see a similar but lower reduction.

When providing both heat and wholesale power, the difference is reduced with SPP performing nearly identically (SI Figure S18). The similarity in performance is due to the dominance of the iMSR across a wide range of facility processes even without the limitation. In ERCOT, we find a similar reduction in HTGRs and similar performance from all other SMR types.



SI Figure S17: Economically viable deployment stacks of SMRs at industrial facility processes in SPP (left) and ERCOT (right) when providing process heat only across avoided natural gas costs. Within each region, the per-facility-process results (from the main text) are in the left column from the per-facility SMR-type limited results are shown in the right column.



SI Figure S18: Economically viable deployment stacks of SMRs at industrial facility processes in SPP (left) and ERCOT (right) when providing process heat and wholesale power across avoided natural gas costs. Within each region, the per-facility-process results (from the main text) are in the left column from the per-facility SMR-type limited results are shown in the right column.

Supplemental References

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4. McMillan, C., and Ruth, M. (2018). Industrial Process Heat Demand Characterization 10.7799/1461488.
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